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A cascade-aware risk index for digital-twin-based, cost-sensitive predictive maintenance in power grids

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ABSTRACT

Modern power grids require maintenance strategies that account not only for component-level fault probability, but also for cascade propagation, operational impact, and intervention cost. This paper proposes the Cascade-Aware Risk Index (CARI), a cost-sensitive, digital-twin-inspired decision-support framework for predictive maintenance, combining fault probability, cascade-propagation propensity, nodal vulnerability, and operational impact to prioritize preventive actions. CARI is evaluated on the IEEE 118-bus system under five severe stress scenarios with measurement noise and partial observability, comparing eight decision policies under common random numbers. Relative to no action, CARI avoids 36,324 MWh-equivalent of blackout-related energy-not-served (ENS) proxy and reduces blackout rate by 0.33, outperforming random (3252 MWh-equivalent; 0.03) and centrality-based (576 MWh-equivalent; 0.00) baselines. Cost-aware CARI preserves nearly the same benefit while lowering the action-cost proxy by 33% and the false-positive intervention cost by 44%. A validation on two additional, independent IEEE benchmarks (39-bus and 57-bus), using the IEEE-118-bus calibration without re-tuning, shows that CARI's risk-ranking mechanism remains structurally consistent across all three networks (Jaccard CARI-vs-random overlap 0.057–0.062), but the calibrated action layer does not transfer without re-calibration: on the 39-bus system every policy except centrality becomes significantly harmful relative to no action, while on the 57-bus system the baseline saturates completely and only fault-probability ranking shows significant improvement. Stress-scenario severity and action thresholds must therefore be re-tuned to the target network before deployment. These results show that combining cascade-aware risk scoring with cost discipline preserves most reliability benefit while reducing unnecessary intervention on a calibrated network.

1. Introduction

The reliable operation of electric power grids has become increasingly challenging due to the combined effects of aging infrastructure, growing electricity demand, renewable energy integration, climate-induced disturbances, and the increasing complexity of distribution and transmission networks. In this context, maintenance is no longer limited to the periodic inspection or replacement of equipment. It has progressively evolved toward condition-based maintenance, risk-informed maintenance, and predictive maintenance strategies that seek to anticipate failures before they propagate into large-scale outages [1–4]. This evolution is particularly important for power utilities operating under technical, financial, and data-quality constraints, where the consequences of delayed intervention may include service interruption, revenue loss, equipment degradation, customer dissatisfaction, and cascading network failures.

Traditional maintenance strategies in power systems have historically relied on corrective maintenance and preventive maintenance. Corrective maintenance is reactive by nature: actions are triggered only after a component has failed. Although simple to implement, this approach may lead to prolonged outages, high emergency repair costs, and poor control over system reliability [1,2]. Preventive maintenance improves upon this by scheduling inspections or replacements according to fixed time intervals or expected equipment lifetimes. However, time-based maintenance may result in unnecessary interventions on healthy assets, while still failing to detect hidden or fast-developing degradation mechanisms [3]. For this reason, modern power-system

Abbreviations: AC, Alternating Current; BR, Blackout Rate; CARI, Cascade-Aware Risk Index; CE_{ENS}, Cost-Effectiveness for ENS reduction; CI, Confidence Interval; CS, Cascade-Size proxy; DT, Digital Twin; ENS, Energy Not Served (proxy); GAT, Graph Attention Network; GNN, Graph Neural Network; IEEE, Inst. of Electrical and Electronics Engineers; MC, Monte Carlo; p.u., per unit; PMU, Phasor Measurement Unit; RCM, Reliability-Centered Maintenance; RL, Reinforcement Learning; SCADA, Supervisory Control and Data Acquisition; SE_{ENS}, Savings-Effectiveness for ENS reduction

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asset management increasingly favors predictive maintenance, where operational measurements, historical failure records, environmental variables, and network states are used to estimate the probability of failure and support early decision-making [4–6].

Recent reviews on predictive maintenance in smart grids emphasize that the transition from conventional maintenance to data-driven maintenance is enabled by the growing availability of sensors, smart meters, SCADA systems, phasor measurement units, fault recorders, and remote monitoring platforms [4,5,7]. The communication and Internet-of-Things (IoT) layer that underpins this sensing infrastructure is itself an active research area: recent reviews comprehensively cover the architectures, protocols, and open challenges of IoT-assisted smart-grid communication [7], while blockchain-coordinated and peer-to-peer grid-digitalization concepts illustrate the broader trend toward decentralized, communication-rich power-system infrastructures [8]. These sources provide large volumes of heterogeneous data related to voltage profiles, current flows, loading levels, breaker states, equipment temperatures, protection events, and customer interruptions. When properly exploited, such data can reveal early signatures of abnormal operating conditions, including overload, undervoltage, insulation degradation, abnormal losses, incipient faults, and cascading vulnerability. Mahmoud et al. [4], for example, provide a systematic review of predictive maintenance in smart grid distribution networks and show that fault prediction requires not only identifying the type and cause of failures, but also understanding the mechanisms through which local disturbances may affect the broader network. This point is essential because a fault in a power grid is rarely an isolated event: its operational impact depends on the location of the fault, the surrounding topology, the loading condition, and the network's ability to absorb or redistribute stress.

Machine learning has therefore become a major tool for power-system monitoring and maintenance. Classical models such as support vector machines, random forests, gradient boosting methods, and ensemble classifiers have been widely used for fault classification, outage prediction, equipment health estimation, and reliability assessment [5,6,9,10]. These models are attractive because they can handle nonlinear relationships and can often perform well even with moderate-sized datasets. However, they generally treat observations as independent feature vectors and do not explicitly represent the physical structure of the power grid. In a networked infrastructure such as an electrical grid, this limitation is critical. The risk associated with a bus, line, transformer, or load point depends not only on its local measurements, but also on the state of neighboring components and on the electrical paths through which disturbances may propagate.

Deep learning has further improved the ability to extract complex temporal and spatial patterns from power-system data. Recurrent neural networks, long short-term memory networks, temporal convolutional networks, and attention-based models have been applied to load forecasting, fault diagnosis, transient stability assessment, anomaly detection, and outage prediction [9–12]. These models are especially useful when historical measurements contain sequential dependencies. For instance, the evolution of voltage magnitudes, loading rates, and fault indicators over time can reveal whether the system is moving toward a stable operating point or toward a degraded state. Nevertheless, purely temporal models remain incomplete for maintenance planning because they generally do not encode the grid topology. A temporal model may detect that a measurement is abnormal, but it may not identify whether the abnormality is located in a structurally critical region, whether neighboring buses are also at risk, or whether a local disturbance is likely to propagate.

To address this limitation, graph-based learning has gained increasing attention in power-system research. Although graph neural networks are reviewed below as conceptual motivation, the cascade-aware index proposed in this paper is a weighted analytic combination of interpretable features rather than a learned graph model; we make this distinction explicit throughout the paper to avoid the impression that a GNN is trained or deployed. Graph neural networks provide a natural framework for representing power grids, where buses are modeled as nodes and transmission or distribution branches are modeled as edges [13–16]. Unlike conventional machine-learning models, GNNs can combine electrical measurements with topological information and learn how risk propagates through the network. Graph convolutional networks, graph attention networks, and their variants have been used for state estimation, contingency screening, load forecasting, fault localization, and stability assessment. Among these models, graph attention mechanisms are particularly relevant because they allow the model to assign different importance levels to neighboring nodes. This is important in power systems because not all neighboring buses contribute equally to risk propagation: the impact of a disturbance depends on electrical proximity, line loading, impedance, network redundancy, and the operating state of connected components. In the present study, graph-based learning is used primarily as a conceptual and methodological motivation: the proposed framework does not train a graph neural network, but it adopts a graph-informed representation of the grid to combine topological vulnerability with electrical stress and cascade-propagation indicators.

In parallel, the concept of the digital twin has emerged as a promising paradigm for power-system monitoring and maintenance. A digital twin can be understood as a dynamic virtual representation of a physical system, continuously updated through data, models, and simulations [17–19]. In power grids, digital twins make it possible to reproduce operating conditions, simulate contingencies, evaluate system vulnerability, test corrective or preventive actions, and support decision-making before interventions are applied in the field. Recent literature on energy digital twins highlights their potential for asset management, reliability assessment, fault detection, operational planning, and resilience improvement [17,18]. However, many existing digital-twin applications remain focused on monitoring and visualization. Their integration with predictive maintenance and intervention planning is still developing, especially when the objective is not only to detect risk, but also to identify where and how preventive actions should be deployed.

Another important gap concerns cascading failures. In power systems, severe outages often result from a sequence of dependent events rather than from a single isolated fault. An initial line outage, overload, or voltage instability may redistribute flows, increase stress on neighboring components, trigger additional trips, and eventually lead to partial or widespread blackout [20,21]. Conventional reliability indices and static centrality measures can provide useful information about structural vulnerability, but they are often insufficient to capture the dynamic nature of cascade propagation. A bus with high topological centrality may not always be the most dangerous under a given operating condition, while a bus with modest centrality may become critical if it is located in a stressed electrical region. This distinction is central to maintenance planning: interventions should not be based solely on structural importance, but on a combined assessment of fault probability, cascade-propagation propensity, nodal vulnerability, and operational impact.

Recent studies increasingly recognize the need for physics-informed and risk-aware artificial intelligence in power systems [22–24]. Instead of relying only on black-box prediction, physics-informed approaches seek to incorporate electrical laws, grid topology, impedance relationships, power-flow constraints, or operational limits into the learning and decision process. This direction is particularly relevant for maintenance applications because utility operators require not only accurate predictions, but also interpretable and actionable indicators. A useful maintenance index should answer practical questions: Which zones are currently at risk? Which components are likely to trigger propagation? Which buses should be monitored, relieved, or reinforced? What is the expected reduction in blackout risk or unserved energy? What is the cost of preventive action compared with the avoided loss?

Despite these advances, three major limitations remain in the existing literature. First, many fault-detection approaches focus on classification accuracy but do not explicitly evaluate the operational consequence of missed faults, such as cascading size, blackout rate, or energy-not-served. Second, several graph-based studies exploit topology but do not sufficiently distinguish between structural centrality and dynamic cascade risk. In real power grids, the most central node is not necessarily the most urgent maintenance target under stressed loading or partial observability. Third, few studies jointly consider risk reduction and intervention cost. This is a critical limitation for practical maintenance planning because utilities operate under budget constraints and cannot intervene everywhere at the same time. A maintenance-oriented decision framework must therefore be able to prioritize actions according to both technical effectiveness and cost efficiency. The novelty of the present work is therefore to transform cascade-risk estimation into a cost-aware maintenance action-ranking problem, rather than treating fault prediction as an isolated classification task.

To address these gaps, this paper proposes a cascade-aware and cost-aware maintenance framework based on a digital-twin benchmark simulation of the IEEE 118-bus power system. The proposed approach introduces a Cascade-Aware Risk Index, denoted CARI, which combines four complementary components: fault probability, cascade-propagation propensity, nodal vulnerability, and operational impact. Unlike static centrality-based prioritization, CARI is designed to capture the dynamic vulnerability of buses under stressed operating conditions, noisy measurements, partial observability, and multi-contingency scenarios. The framework evaluates several decision policies, including no action, random selection, centrality-based selection, fault-probability-based selection, condition-based maintenance, scheduled maintenance, cascade-aware CARI, and cost-aware CARI. Preventive and mitigation actions are modeled through regional load relief, reactive support, and targeted thermal-limit enhancement proxies, while their operational costs are explicitly accounted for.

The main contribution of this study is therefore not limited to fault prediction. Rather, it provides a demonstrative decision-support framework for predictive maintenance under cascade risk. The proposed method links digital-twin benchmark simulation, graph-based risk representation, cascade-aware prioritization, and cost-sensitive intervention assessment. Simulation results show that CARI-based policies substantially reduce cascade severity, blackout-related energy-not-served (ENS) proxy, and blackout rate compared with no-action, random, and centrality-based baselines. In particular, the cost-aware CARI policy achieves comparable blackout-risk reduction to the full cascade-aware policy while requiring lower action cost and lower planned load shedding. This makes the framework especially relevant for utilities seeking to improve reliability while maintaining operational and financial feasibility.

The specific contributions of this paper can be summarized as follows.

- A Cascade-Aware Risk Index (CARI) combining fault probability, cascade-propagation propensity, nodal vulnerability, and operational impact into a single, interpretable, weighted prioritization score for predictive maintenance.
- A cost-aware variant of CARI that preserves nearly the full avoided-ENS and blackout-rate benefit of standard CARI while cutting the average action-cost proxy by about 33% (33.4 vs. 50.2) and the false-positive intervention burden by about 44% (28.6 vs. 51.0).
- A digital-twin-inspired benchmark environment on the IEEE 118-bus system jointly emulating stressed loading, multi-contingency outages, measurement noise, and partial observability across five representative disturbance mechanisms.
- A systematic comparison of eight decision policies – no action, random, centrality, fault-probability, condition-based, scheduled, CARI, and cost-aware CARI – under common random numbers, enabling matched statistical comparison.
- Diagnostic analyses (Jaccard overlap between policies, high-vulnerability/low-centrality bus detection, cost-versus-benefit Pareto frontiers) demonstrating that CARI captures dynamic, operationally driven risk that purely structural centrality misses.

Table 1 positions the present work relative to the main streams of the existing literature reviewed above.

The remainder of this paper is organized as follows. Section 2 presents the proposed cascade-aware digital-twin methodology, including the benchmark grid and power-flow model, the graph-based topological indicators, the stress-scenario and contingency-generation procedure, the electrical-stress and fault-probability indicators, the Cascade-Aware Risk Index (CARI) and its cost-aware variant, the decision policies and persistent mitigation actions, the action-cost model, and the evaluation metrics and statistical-comparison procedure. Section 3 reports and discusses the simulation results, including overall policy performance, scenario-wise reduction patterns, marginal cost-effectiveness, spatial interpretation of CARI-based decisions, and the false-positive operational burden. Section 4 concludes the paper, summarizes its main limitations, and outlines directions for future work.

2. Materials and methods

2.1. General methodological framework

This study proposes a cost-aware cascade-risk mitigation framework for power-system operation under severe stress conditions. The objective is not only to detect vulnerable buses, but also to evaluate whether preventive or corrective actions selected from a risk index can reduce cascading effects, blackout-related energy-not-served (ENS) proxy, and operational risk at an acceptable intervention cost. The proposed methodology is built around a digital-twin simulation environment, where the IEEE 118-bus benchmark system is used as a controlled test grid. The IEEE 118-bus system remains widely adopted in power-system studies because it provides a sufficiently complex meshed network, while remaining computationally tractable for repeated contingency and decision-policy simulations [32,33]. Specifically, as instantiated through `pandapower`'s standard `case118` model, the benchmark comprises 118 buses, 186 branches (177 lines and 9 transformers), 91 loads, and 54 generators, forming a meshed topology with multiple alternative paths between most bus pairs — a property that is necessary for flow redistribution and cascade propagation to be meaningfully studied, unlike a radial or sparsely connected network.

The simulation framework combines four main components: first, an electrical network model solved by Newton–Raphson power flow; second, a graph representation of the grid topology; third, a nodal cascade-risk scoring mechanism; and fourth, a decision layer that converts risk scores into preventive or mitigation actions. The electrical simulations are implemented in pandapower, an open-source Python tool designed for power-flow analysis, optimal power flow, state estimation, short-circuit studies, and topological analyses of electric power systems [34]. The graph-based representation is constructed using NetworkX, which allows the extraction of structural indicators such as degree centrality, betweenness centrality, and clustering coefficients [35]. These indicators are not used alone as the final decision criterion; rather, they are combined with electrical stress, load impact, fault probability, and cascade-propagation propensity in a composite index called the Cascade-Aware Risk Index (CARI).

The general workflow is summarized as follows. The IEEE 118-bus network is first loaded and converted into a graph. Then, for each stress scenario and Monte Carlo run, active and reactive loads are scaled to emulate severe operating conditions. Initial line outages are introduced at selected critical lines to trigger possible cascading effects. At each simulation step, the power flow is solved and nodal risk features are computed. Depending on the policy under evaluation, a subset of buses

Table 1
Positioning of the proposed framework relative to existing literature streams.

Literature stream	Topology-aware	Models cascade dynamics	Cost/intervention-aware	Validation basis
Classical reliability/RCM [1–3,25]	No	No	Yes (asset-level)	Analytical/utility data
Topological vulnerability studies [20,26–28]	Yes	Partial (static)	No	Benchmark/synthetic
ML/DL fault and outage prediction [4,6,10]	No	No	No	Historical/utility data
GNN/GAT power-system studies [15,29,30]	Yes	Partial (learned)	No	Simulation/benchmark
RL-based remedial actions [31]	Yes (implicit)	Yes	Partial (via reward)	IEEE 39-/57-bus simulation
Digital-twin surveys [17–19]	n/a	n/a	n/a	Conceptual/review
This study (CARI/cost-aware CARI)	**Yes**	**Yes**	**Yes**	**IEEE 118-bus, 5 scenarios, Monte Carlo**

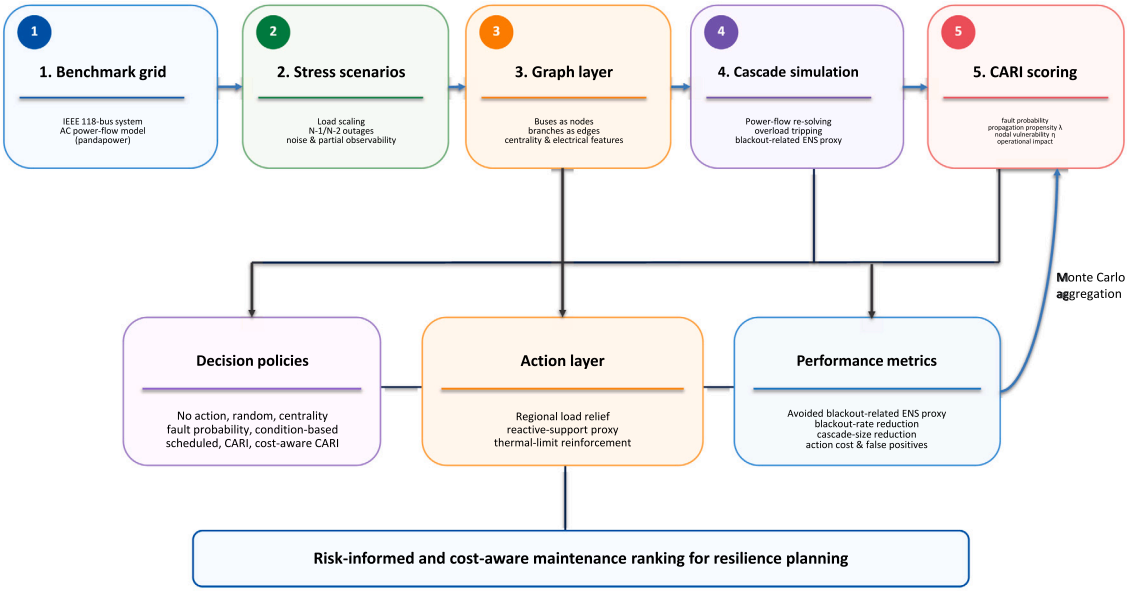


Fig. 1. Methodological workflow of the proposed cascade-aware digital twin framework. The pipeline links benchmark-grid simulation, stress-scenario generation, graph-based representation, cascading-event simulation, CARI-based prioritization, cost-aware action selection, and resilience-oriented performance evaluation.

is selected for action. The action layer then applies persistent interventions such as regional load relief, reactive-support proxy, and local or global line-capacity reinforcement. The system is subsequently re-solved to assess whether the selected policy reduces cascading events, voltage stress, blackout occurrence, and blackout-related ENS proxy.

Fig. 1 summarizes this end-to-end workflow and clarifies the links between the simulation environment, the graph-based representation, CARI computation, decision-policy selection, intervention modeling, and performance assessment. Algorithm 1 restates the same workflow at the level of pseudocode, for reproducibility.

2.2. Scope and limitations of the digital-twin representation

In this paper, the term *digital twin* denotes a physics-based virtual replica of the IEEE 118-bus system – solved through AC power flow (Section 2.3) and combined with a graph representation (Section 2.4) – that can be subjected to scenario, contingency, and intervention testing before any action is considered for the physical system. This satisfies the core digital-twin requirement of a model-based virtual environment used to evaluate interventions safely offline. However, the present implementation does not include two features often associated with a fully operational digital twin: (i) continuous, automated synchronization of the virtual model's state with live field measurements (the noisy/partially observable measurements in Section 2.6 are simulated, not ingested from an operational SCADA/PMU system), and (ii) closed-loop, bidirectional interaction in which actions computed by the model are automatically dispatched to, and confirmed by, physical grid assets. The framework is therefore more precisely described as a *digital-twin-inspired benchmark simulation*, used here to evaluate decision policies under repeatable, controlled conditions rather than as a deployed, real-time digital twin. This distinction is revisited in the Limitations subsection (Section 3.9), while the underlying physics-based simulation and decision layer are directly reusable as components of a future real-time digital twin once live data integration is added.

2.3. Digital twin test system and power-flow model

Let the power grid be represented by a set of buses

$$\mathcal{N} = \{1, 2, \ldots, N\} \tag{1}$$

and a set of branches

$$\mathcal{E} = \{(i, j) : i, j \in \mathcal{N}\} \tag{2}$$

In this study, $N = 118$, corresponding to the IEEE 118-bus benchmark. Each bus i is characterized by its voltage magnitude V_i, active load demand P_i, reactive load demand Q_i, and its topological position in the network. Each branch (i, j) is characterized by its series impedance

$$z_{ij} = r_{ij} + jx_{ij} \tag{3}$$

where r_{ij} and x_{ij} are the branch resistance and reactance, respectively. The corresponding admittance magnitude is given by

$$|y_{ij}| = \left| \frac{1}{z_{ij}} \right| \tag{4}$$

Algorithm 1 CARI-based cascade-aware decision loop

Require: grid model $G = (\mathcal{N}, \mathcal{E})$; scenario set Ω; policy π; number of Monte Carlo runs R; horizon T; action-set size K

for each scenario $s \in \Omega$ **do**
  for each run $r = 1, \dots, R$ **do**
    Initialize base case; apply scenario-specific load scaling (Eqs. (13)–(14))
    Draw common random numbers (load noise, measurement noise/missing mask, fault noise, line-trip draws), shared across all policies
    Inject initial line outage(s) for scenario s at the decision time immediately following $t = 1$
    for each time step $t = 1, \dots, T$ **do**
      Solve AC power flow (Newton–Raphson); if non-convergent: flag blackout, end run
      Compute U_i, U_i^{sev}, O_i, C_i (Eqs. (18)–(9))
      Compute P_i^{fault}, λ_i, η_i, I_i, $CARI_i$ (Eqs. (22)–(29))
      if π is cost-aware **then**
        Compute S_i^{CA} (Eq. (34))
      end if
      if t is a decision time **then**
        Select action set $A_t^{\pi} = \text{TopK}(S_i^{\pi})$ (Eq. (35))
        Apply persistent actions (load relief, reactive support, thermal-limit reinforcement; Eqs. (37)–(40)) subject to $\Gamma_{\max}$, $\Delta_{Q,\max}$, and single-boost-per-line constraints
        Re-solve power flow
      end if
      Evaluate overloaded-line trip probabilities (Eqs. (43)–(44)); apply trips; re-solve if needed
      Update cascade-size and ENS-proxy accumulators (Eqs. (45)–(48))
    end for
    Record run-level metrics: CS, ENS_{proxy}, blackout indicator, C^{act}
  end for
  Aggregate scenario-level means, standard deviations, and 95% CI (Eq. (58)) and paired differences vs. π_0 (Eq. (59))
end for

Ensure: per-policy, per-scenario performance table and diagnostic indices (Jaccard, Pareto frontier)

At each simulation time step, the AC power-flow problem is solved using the Newton–Raphson method. The active and reactive power-balance equations at bus i can be expressed as

$$P_i^{\text{inj}} = V_i \sum_{j=1}^{N} V_j \left(G_{ij} \cos\theta_{ij} + B_{ij} \sin\theta_{ij}\right),$$
$$Q_i^{\text{inj}} = V_i \sum_{j=1}^{N} V_j \left(G_{ij} \sin\theta_{ij} - B_{ij} \cos\theta_{ij}\right). \tag{5}$$

where P_i^{inj} and Q_i^{inj} denote the net active and reactive power injections at bus i, G_{ij} and B_{ij} are the conductance and susceptance elements of the bus-admittance matrix, and $\theta_{ij} = \theta_i - \theta_j$ is the voltage-angle difference between buses i and j. A power-flow solution is considered feasible when the Newton–Raphson solver converges within the specified tolerance. If the solver fails to converge after a severe contingency or cascading event, the corresponding state is treated as a blackout condition in the simulation.

The IEEE 118-bus model is used as a demonstrative digital twin rather than as a representation of a specific real utility network. This choice makes it possible to evaluate the proposed decision policy under repeatable and transparent conditions. The use of a benchmark system also facilitates comparison with previous studies on cascading failures, vulnerability assessment, and graph-based power-system analysis [20,26,28,32,33,36].

2.4. Graph representation and topological indicators

The electrical network is converted into an undirected graph

$$G = (\mathcal{N}, \mathcal{E}) \tag{6}$$

where buses are represented as nodes and transmission lines or transformers are represented as edges. For each branch connecting buses i and j, an edge is added to the graph. This graph representation allows the extraction of classical structural indicators that are frequently used in vulnerability and cascading-failure studies [20,26,35].

The degree centrality of a bus i is defined as

$$C_D(i) = \frac{d_i}{N-1} \tag{7}$$

where d_i is the number of adjacent buses connected to bus i. Degree centrality provides a first indication of local connectivity. However, a bus with high degree is not necessarily the most critical from a cascading-risk perspective, because electrical stress also depends on power flows, voltage levels, loading conditions, and the redistribution of flows after contingencies.

The betweenness centrality of bus i is defined as

$$C_B(i) = \sum_{\substack{s,t \in \mathcal{N} \\ s \neq t,\ s \neq i,\ t \neq i}} \frac{\sigma_{st}(i)}{\sigma_{st}} \tag{8}$$

where σ_{st} is the number of shortest paths between buses s and t, and $\sigma_{st}(i)$ is the number of those paths passing through bus i. Betweenness centrality captures the extent to which a bus lies on important topological corridors. In cascading-failure studies, such nodes may play an important role in redistributing power flows after line outages [20,26–28].

A normalized composite centrality score is then computed as

$$C_i = 0.5\bar{C}_D(i) + 0.5\bar{C}_B(i) \tag{9}$$

where $\bar{C}_D(i)$ and $\bar{C}_B(i)$ denote min–max normalized degree and betweenness centrality, respectively. The min–max normalization of a variable x_i is defined as

$$\bar{x}_i = \frac{x_i - \min(x)}{\max(x) - \min(x) + \epsilon} \tag{10}$$

where ϵ is a small numerical constant, set to 10^{-6} in the simulations, introduced to avoid division by zero.

The clustering coefficient $C_{cl}(i)$ is also computed to quantify the local redundancy around a bus. It is defined, as usual, as

$$C_{cl}(i) = \frac{2T_i}{d_i(d_i - 1)} \tag{11}$$

where T_i is the number of triangles through bus i and d_i is its degree. A lower clustering value may indicate reduced local meshing and therefore a weaker capacity to absorb local disturbances. In the proposed model, clustering is not interpreted as a standalone risk indicator. Instead, it is later introduced through the vulnerability term as a proxy for local redundancy: weakly clustered areas are assumed to have fewer alternative paths for absorbing disturbances.

2.5. Stress scenarios and contingency generation

To test the behavior of the decision policies under different operating conditions, five severe scenarios are considered:

$$S = \{S_1, S_2, S_3, S_4, S_5\} \tag{12}$$

corresponding respectively to overload cascade, voltage collapse, line-outage propagation, mixed multifault stress, and noisy partial observability. Each scenario is characterized by active-load scaling,

Table 2
Stress-scenario parameterization used in the implemented simulation.

Scenario	$\alpha_P^{(s)}$	$\alpha_Q^{(s)}$	Initial outages	σ_s	ρ_s
Overload cascade	1.55	1.25	1	0.006	0.05
Voltage collapse	1.30	1.90	1	0.006	0.05
Line-outage propagation	1.40	1.25	2	0.006	0.05
Mixed multi-fault	1.55	1.70	3	0.010	0.08
Noisy partial observability	1.45	1.40	2	0.030	0.20

reactive-load scaling, initial line outages, measurement noise, and missing-data rate. Table 2 reports the exact parameterization used in the implemented simulation. The overload-cascade and mixed multi-fault scenarios apply the most severe active-load scaling, the voltage-collapse scenario applies the most severe reactive-load scaling, the line-outage-propagation scenario relies on multiple topologically targeted initial outages under comparatively moderate loading, and the noisy partial-observability scenario combines severe loading with the highest measurement-noise standard deviation and missing-data rate among all scenarios.

At each time step t, active and reactive loads are scaled according to

$$P_{l,t}^{(s)} = P_{l,0}\alpha_P^{(s)} d_t \xi_t, \quad Q_{l,t}^{(s)} = Q_{l,0}\alpha_Q^{(s)} d_t \xi_t. \tag{13}$$

where $P_{l,0}$ and $Q_{l,0}$ are the base active and reactive loads, $\alpha_P^{(s)}$ and $\alpha_Q^{(s)}$ are the active and reactive scaling factors for scenario s, d_t is a synthetic daily load variation factor, and ξ_t is a stochastic perturbation drawn from a normal distribution centered around 1. In the implemented simulation, the daily factor is given by

$$d_t = 0.92 + 0.25 \sin\left(\frac{2\pi(t-5)}{24}\right) \tag{14}$$

The coefficients in Eq. (14) define a stylized diurnal load profile (a ±25% swing around a 0.92 baseline) used for illustrative scenario generation; they are not fitted to a specific utility load curve. In the implemented simulation, decisions are taken at three points within the horizon: immediately after load scaling and before the forced contingency ($t = 1$), immediately after the forced initial outage ($t = 2$), and at a mid-horizon checkpoint $t = \max(3, \lfloor T/3 \rfloor)$ (i.e., $t = 6$ for the base-experiment horizon $T = 18$), so that cascading scenarios are not reduced to a single binary pre/post-outage outcome.

Initial outages are introduced at critical lines selected from topological scores. Specifically, lines connected to buses with high composite centrality are ranked, and the most critical lines are used as initial contingencies. This choice is consistent with the idea that highly connected or structurally important corridors may accelerate cascade propagation when lost under stressed operating conditions [20,26,28,36].

2.6. Measurement noise and partial observability

To emulate imperfect field measurements, especially under SCADA-like conditions, voltage measurements are corrupted by Gaussian noise and missing observations. Let $V_i^{true}(t)$ be the true simulated voltage magnitude at bus i and time t. The measured voltage is modeled as

$$V_i^{\text{meas}}(t) = V_i^{\text{true}}(t) + \varepsilon_i(t) \tag{15}$$

where

$$\varepsilon_i(t) \sim \mathcal{N}(0, \sigma_s^2) \tag{16}$$

and σ_s is the scenario-dependent measurement-noise standard deviation. Missing measurements are introduced with probability ρ_s. If a measurement is missing, the voltage value is replaced by a neutral fallback value of 1.0 p.u.:

$$V_i^{\text{obs}}(t) = \begin{cases} V_i^{\text{meas}}(t), & \text{if measurement is available,} \\ 1.0, & \text{otherwise.} \end{cases} \tag{17}$$

This neutral fallback represents a conservative imputation used when no reliable alarm information is available from the measurement channel. It may mask some local voltage degradation, but this effect is explicitly tested through the noisy partial-observability scenario. This design makes the simulation more realistic than a fully observable deterministic setting and allows the evaluation of whether the proposed CARI-based decision mechanism remains meaningful when the operator has access only to noisy and partially incomplete nodal measurements.

2.7. Electrical stress indicators

The first electrical indicator is the undervoltage severity. For each bus i, it is computed from the observed voltage as

$$U_i = \max\left(0, \frac{V_{\text{warn}} - V_i^{\text{obs}}}{\Delta V}\right) \tag{18}$$

where $V_{\text{warn}} = 0.97$ p.u. is the warning voltage threshold and $\Delta V = 0.08$ is a scaling factor. The value is clipped between 0 and 1. A second, more severe undervoltage indicator is defined as

$$U_i^{\text{sev}} = \max\left(0, \frac{V_{\text{under}} - V_i^{\text{obs}}}{\Delta V}\right) \tag{19}$$

where $V_{\text{under}} = 0.94$ p.u. This term captures stronger voltage degradation. V_{warn} and V_{under} are deliberately set slightly inside typical steady-state secondary-voltage operating bands, so that developing voltage stress is flagged for ranking purposes before it reaches emergency levels; they are operational ranking conventions used to compare decision policies under identical conditions, rather than protection-system settings.

The second electrical indicator is the overload exposure of each bus. If L_{ij} denotes the loading percentage of line (i, j), the overload contribution of this line is defined as

$$O_{ij} = \min\left(1, \max\left(0, \frac{L_{ij} - 80}{70}\right)\right) \tag{20}$$

The nodal overload exposure is then obtained by summing the overload contributions of all incident lines:

$$O_i = \sum_{j:(i,j)\in\mathcal{E}} O_{ij} \tag{21}$$

The resulting vector is normalized using min–max normalization. The upper clipping prevents extremely overloaded branches from dominating the composite score disproportionately. The voltage and loading thresholds used in Eqs. (18)–(20) are operational stress-ranking thresholds rather than universal protection settings. They are used to compare policies under the same simulated conditions. This indicator is important because line overloads are one of the main mechanisms through which cascading failures propagate in stressed transmission networks [20,26,36].

2.8. Fault-probability proxy

A nodal fault-probability proxy is computed by combining voltage stress, overload exposure, load level, centrality, and reactive demand. Throughout this and the following subsections, $\bar{P}_i$ and $\bar{Q}_i$ denote the active and reactive load levels P_i and Q_i normalized via the min–max transform of Eq. (10). In the implementation, the raw fault score is defined as

$$z_i = 2.50U_i + 2.00O_i + 0.60\bar{P}_i + 0.45C_i + 0.25\bar{Q}_i - 1.25 + \epsilon_i \tag{22}$$

where U_i is the undervoltage severity, O_i is the overload exposure, $\bar{P}_i$ and $\bar{Q}_i$ are normalized active and reactive load levels, C_i is the composite centrality score, and $\epsilon_i \sim \mathcal{N}(0, 0.07^2)$ is a small, independent per-bus stochastic disturbance (distinct from the measurement noise $\varepsilon_i(t)$ of Eq. (15)). The nodal fault probability is then obtained through a sigmoid transformation:

$$P_i^{\text{fault}} = \frac{1}{1 + \exp(-z_i)} \tag{23}$$

The coefficients in Eq. (22) were selected as heuristic stress-ranking weights after preliminary sensitivity tests, with dominant emphasis placed on undervoltage severity and overload exposure because these two variables are the most direct indicators of degraded operating conditions in the simulated cascade process. Load level, reactive demand, and structural vulnerability are retained as secondary factors. This formulation does not claim to be a calibrated protection-system failure probability or to represent physical failure rates. Rather, it provides a consistent normalized risk proxy for comparing decision policies under identical simulation conditions and identifying buses where electrical stress and structural vulnerability coincide.

2.9. Cascade-propagation propensity

The cascade-propagation propensity, denoted by λ_i, measures the likelihood that a disturbance around bus i may contribute to further propagation. It is inspired by the idea that cascading events are not only local failures, but dynamic processes where overloads, voltage degradation, and network structure interact [20,26,28,36]. In the proposed framework, λ_i is defined as

$$\lambda_i = 0.48P_i^{\text{fault}} + 0.30O_i + 0.15U_i^{\text{sev}} + 0.07C_i \tag{24}$$

The term P_i^{fault} captures the local probability of abnormal operation, O_i reflects flow stress, U_i^{sev} captures severe voltage weakness, and C_i accounts for structural importance. The resulting value is clipped to the interval $[0, 1]$. In this study, λ_i is not a stochastic event intensity in the point-process sense; it is a normalized cascade-propagation score used for ranking maintenance actions.

2.10. Nodal vulnerability and operational impact

The vulnerability factor η_i measures the propensity of a bus to contribute to system degradation when subjected to electrical stress. In the implemented model, it is computed as

$$\eta_i = 0.32P_i^{\text{fault}} + 0.18C_i + 0.35O_i + 0.10U_i^{\text{sev}} + 0.05(1 - C_{cl}(i)) \tag{25}$$

This formulation gives more weight to overload exposure, because overloaded branches directly contribute to cascading line trips. It also includes the inverse of the clustering coefficient to capture the fact that weakly clustered areas may have fewer local alternative paths.

The operational impact of bus i is defined as

$$I_i = 0.55\bar{P}_i + 0.25C_i + 0.20\bar{Q}_i \tag{26}$$

This term reflects the fact that intervention priority should not depend only on the probability of a disturbance, but also on the possible consequence of that disturbance. A bus supplying large active and reactive demand, or occupying an important structural position, may have a higher operational impact if affected. Conceptually, λ_i captures the immediate propensity of a local disturbance to propagate, η_i captures the broader nodal vulnerability under electrical stress, and I_i captures the expected operational consequence. This distinction is important because the three terms share some input variables but represent different decision dimensions.

2.11. Cascade-aware risk index

The central methodological contribution is the construction of the Cascade-Aware Risk Index:

$$CARI_i = w_f P_i^{\text{fault}} + w_\lambda \lambda_i + w_\eta \eta_i + w_I I_i \tag{27}$$

The weights used in the base configuration are

$$w_f = 0.20, \quad w_\lambda = 0.30, \quad w_\eta = 0.35, \quad w_I = 0.15 \tag{28}$$

The weights are normalized such that $w_f + w_\lambda + w_\eta + w_I = 1$.

Thus,

$$CARI_i = 0.20P_i^{\text{fault}} + 0.30\lambda_i + 0.35\eta_i + 0.15I_i \tag{29}$$

The weighting scheme intentionally gives higher importance to λ_i and η_i, because the purpose of the framework is not only to detect local abnormality, but to identify buses that may contribute to cascade propagation. This distinguishes the proposed index from purely topological selection rules or simple fault-probability ranking. A bus with moderate centrality can still obtain a high CARI value if it is located in an electrically stressed region, has high overload exposure, or shows strong voltage vulnerability.

For the ablation analysis, four reduced versions of CARI are also computed:

$$CARI^{-\eta}, \quad CARI^{-\lambda}, \quad CARI^{-I}, \quad CARI^{-f} \tag{30}$$

Each reduced index removes one component and renormalizes the remaining weights. This allows the contribution of vulnerability, propagation propensity, impact, and fault probability to be assessed separately.

2.11.1. Redundancy among CARI components

Because λ_i and η_i are deliberately built by reusing P_i^{fault} and the overload exposure O_i (Eqs. (24)–(25)), the four CARI components are not statistically independent. Table S1 of the Supplementary Material reports the Pearson correlation among P_i^{fault}, λ_i, η_i, and I_i, computed from a diagnostic snapshot of all 118 buses under each of the five stress scenarios (590 bus-scenario observations; pre-contingency, i.e. before the forced initial outage at $t = 2$). The correlation between P_i^{fault} and λ_i is very high ($r = 0.95$; Spearman $\rho = 0.94$) and is stable across all five scenarios individually (r between 0.91 and 0.98), confirming that this overlap is a structural property of the formulation rather than a sample artifact. η_i is moderately correlated with both ($r = 0.61$ with P_i^{fault}, $r = 0.81$ with λ_i), while I_i is the least redundant component ($r \leq 0.55$ with the other three). A variance-inflation-factor (VIF) analysis of the four components as linear predictors confirms severe multicollinearity for P_i^{fault} (VIF = 31.2) and λ_i (VIF = 56.2), moderate multicollinearity for η_i (VIF = 9.0), and comparatively low multicollinearity for I_i (VIF = 1.5); a principal component analysis of the same four standardized components shows that a single component already explains 72.8% of their joint variance (95.0% for the first two components).

This redundancy is by design – λ_i and η_i are intended as higher-order constructs that build on the local fault signal and on overload exposure, in the same way that many composite reliability indices reuse a base likelihood term in downstream products [25] – but it does mean that CARI's four nominal "components" do not contribute four independent pieces of information. The ablation indices of Eq. (30) should therefore be interpreted with this overlap in mind: removing λ_i or η_i does not remove the fault-probability signal itself, since it re-enters CARI both directly ($w_f P_i^{\text{fault}}$) and indirectly through whichever of λ_i, η_i remains. The comparatively low collinearity of I_i confirms that operational impact is the component contributing the most genuinely independent information to CARI.

2.12. Decision policies

Eight decision policies are compared:

$$\Pi = \{\pi_0, \pi_R, \pi_C, \pi_F, \pi_{CARI}, \pi_{CA\text{-}CARI}, \pi_{CB}, \pi_{SCH}\} \tag{31}$$

Throughout this paper, π denotes a decision policy from the set Π, not the mathematical constant.

The no-action policy π_0 serves as the corrective-only baseline. The random policy π_R selects buses uniformly at random and is used to verify that performance gains are not caused by intervention volume alone. The centrality policy π_C selects buses with the highest composite topological centrality defined in Eq. (9). The fault-probability policy π_F selects buses according to the local risk proxy P_i^{fault} defined in Eq. (23).

The cascade-aware policy π_{CARI} selects buses with the highest CARI values defined in Eq. (29).

Two operational maintenance baselines are also included. The condition-based policy π_{CB} selects buses according to a purely local electrical-stress score,

$$\mathrm{cond}_i = 0.50O_i + 0.30U_i^{\mathrm{sev}} + 0.20\bar{P}_l, \tag{32}$$

and applies a preventive action when $\mathrm{cond}_{prevent} \le \mathrm{cond}_i < \mathrm{cond}_{mitigate}$ and a mitigation action above $\mathrm{cond}_{mitigate}$, using thresholds $\mathrm{cond}_{prevent} = 0.12$ and $\mathrm{cond}_{mitigate} = 0.35$ and relief/shedding/reactive-support magnitudes of 0.06, 0.10, and 0.08, respectively. Unlike CARI, cond_i does not include centrality, fault probability, or propagation propensity: it represents a conventional, locally-triggered condition-monitoring rule. The scheduled maintenance policy π_{SCH} instead ranks buses once, before any scenario is run, using a fixed priority score computed only from nominal loading and topology,

$$\mathrm{sched}_i = 0.60\left(0.70\bar{P}_i^0 + 0.30\bar{Q}_i^0\right) + 0.40C_i, \tag{33}$$

where $\bar{P}_i^0$, $\bar{Q}_i^0$ are normalized nominal (pre-scenario) active and reactive loads. The top-K buses under sched_i unconditionally receive a preventive action (relief and reactive-support magnitude 0.04 each, plus local line-capacity reinforcement) at every decision time, regardless of the instantaneous network state — consistent with periodic, schedule-driven preventive maintenance that does not respond to real-time risk. These two additional policies allow the proposed adaptive CARI policies to be compared with commonly used maintenance logics: threshold-driven condition-based maintenance and periodic preventive maintenance.

The cost-aware CARI policy $\pi_{CA\text{-}CARI}$ uses a modified score:

$$S_i^{CA} = CARI_i + 0.15\eta_i + 0.10O_i - 0.03I_i \tag{34}$$

This formulation keeps the main cascade-risk information while slightly penalizing high-impact load-shedding zones, unless their risk is sufficiently high. The negative coefficient applied to I_i in Eq. (34) does not mean that operational impact is unimportant; rather, it reflects the additional intervention burden associated with acting on high-load zones when the incremental cascade-risk benefit is limited. In other words, the cost-aware policy aims to preserve most of the system-protection benefit of CARI while reducing unnecessary intervention cost.

In addition, under scenarios with substantial measurement degradation – specifically, when the scenario-level measurement-noise standard deviation satisfies $\sigma_s \ge 0.02$ or the missing-data rate satisfies $\rho_s \ge 0.15$ (which, among the five scenarios of Table 2, applies only to the noisy partial-observability scenario) – the cost-aware policy additionally suppresses the costly preventive load-relief, mitigation-shedding, and reactive-support actions (setting their magnitudes to zero), while still permitting the comparatively cheap local and global thermal-limit reinforcement. The rationale is that committing to costly, hard-to-reverse load actions is not justified when the underlying CARI score is itself computed from unreliable measurements. This design choice directly explains why, under the noisy partial-observability scenario, the cost-aware policy is more conservative than standard CARI in Table 7 (Section 3.2): the discrepancy reflects an intentional cost-aware safeguard against acting confidently on noisy information, rather than a weakness of the underlying risk index.

For each decision time, the selected bus set is

$$A_t^{\pi} = \mathrm{TopK}\left(S_i^{\pi}\right) \tag{35}$$

where S_i^{π} is the ranking score associated with policy π, and K is the number of buses selected for action. In the base experiment, $K = 12$ for most policies, while the cost-aware policy uses a smaller effective action set, $K_{CA} = 8$, reflecting its more selective intervention philosophy.

Selection by Eq. (35) determines *which* buses are eligible for action; a second, threshold-based stage then determines *which type* of action, if any, is actually applied to each selected bus. For a selected bus with score S_i^{π} (or $CARI_i$ for non-cost-aware policies), three thresholds delineate the action tiers:

$$\theta_{watch} = 0.25, \quad \theta_{prevent} = 0.34, \quad \theta_{mitigate} = 0.52 \tag{36}$$

A selected bus below θ_{watch} receives no action; a bus with score in $[\theta_{watch}, \theta_{prevent})$ receives monitoring only; a bus with score in $[\theta_{prevent}, \theta_{mitigate})$ receives a preventive action (local line-capacity reinforcement and regional active/reactive-load relief); and a bus at or above $\theta_{mitigate}$ receives a mitigation action (the preventive action plus an additional reactive-support proxy, and a one-time global reinforcement of the most loaded lines is triggered the first time any bus in the decision step crosses $\theta_{prevent}$). For the cost-aware policy, θ_{watch} is set equal to $\theta_{prevent}$ (both = 0.34), so that the monitoring-only tier is skipped entirely and cost is incurred only when the cost-adjusted score already justifies a physical action; $\theta_{prevent}$ and $\theta_{mitigate}$ are otherwise unchanged. This threshold structure, together with the TopK pre-selection of Eq. (35), is what ultimately determines the action-cost values reported in Section 3.

2.13. Persistent preventive and mitigation actions

The decision layer translates selected buses into simplified physical or operational intervention proxies. Three types of actions are modeled. They should be interpreted as controllable mitigation abstractions rather than detailed field-operation models.

First, regional active-load relief is applied around selected buses. For a load l connected to a selected bus or one of its neighboring buses, the persistent active-load reduction is

$$P_l^{\mathrm{new}} = \max\left[P_l(1-\gamma), P_{l,0}(1-\Gamma_{\max})\right] \tag{37}$$

where γ is the action-dependent relief factor and $\Gamma_{\max} = 0.35$ is the maximum cumulative planned-shedding fraction. In the implementation, $\gamma = 0.08$ for a preventive action and $\gamma = 0.15$ for a mitigation action ($\gamma = 0.08$ for both action tiers under the cost-aware policy, except where suppressed entirely as described in Section 2.12). This cap prevents unrealistic repeated shedding of the same load. The reactive demand is reduced in the same proportion:

$$Q_l^{\mathrm{new}} = \max\left[Q_l(1-\gamma), Q_{l,0}(1-\Gamma_{\max})\right] \tag{38}$$

Second, a reactive-support proxy is implemented as a reduction in local reactive demand:

$$Q_l^{\mathrm{new}} = \max\left[Q_l(1-\delta_Q), Q_{l,0}(1-\Delta_{Q,\max})\right] \tag{39}$$

where δ_Q is the reactive-support factor and $\Delta_{Q,\max} = 0.35$ is the maximum cumulative reactive-support fraction. In the implementation, $\delta_Q = 0.10$ under the standard mitigation tier and $\delta_Q = 0.08$ under the cost-aware policy (again subject to the suppression rule of Section 2.12 under noisy partial observability).

Third, local and global thermal-limit enhancement proxies are modeled by increasing the effective current limit of selected lines. From this point on, a branch is indexed generically by a single edge index e (with $e = (i, j)$ for the branch connecting buses i and j), rather than by the bus-pair subscript (i, j) used in Section 2.7; both notations refer to the same underlying branch set $\mathcal{E}$. This represents an abstracted mitigation effect, such as temporary operational relief, dynamic rating, or local reinforcement planning. If line e is reinforced, its current limit becomes

$$I_e^{\max,\mathrm{new}} = \beta I_e^{\max} \tag{40}$$

where $\beta = 1.15$ for local incident-line reinforcement and $\beta = 1.25$ for global reinforcement of the most loaded lines. Each line can be boosted only once per simulation run. This guard is important because repeated artificial capacity boosts would otherwise exaggerate the effectiveness of intervention policies.

2.14. Action-cost model

Each intervention is associated with a cost proxy. The total action cost for a simulation run is

$$C_{act} = c_m N_m + c_p N_p + c_g N_g + c_s P_{shed} + c_q Q_{\text{sup}} \tag{41}$$

where N_m, N_p, and N_g are the numbers of monitoring, preventive, and mitigation actions, respectively. P_{shed} is the total planned active-load relief, and Q_{sup} is the reactive-support proxy. The unit costs used in the base configuration are

$$\begin{aligned} &c_m = 0.25, \quad c_p = 1.00, \quad c_g = 2.00, \\ &c_s = 0.15, \quad c_q = 0.05. \end{aligned} \tag{42}$$

The purpose of this cost model is not to reproduce actual utility cost accounting. It is used as a normalized operational proxy to compare policies under the same assumptions. The unit costs in Eq. (42) represent relative intervention burdens rather than monetary values, ordered as follows: monitoring (c_m) is the least resource-intensive action and is therefore weighted lowest; load shedding and reactive support (c_s, c_q) are simple, fast, per-unit operational actions; preventive line-level actions (c_p) require crew dispatch and switching and are therefore weighted higher; and global reinforcement (c_g) is the most resource-intensive, multi-line planning action and is weighted highest. This is especially important because a risk-reduction method that requires excessive preventive shedding may be operationally unattractive even if it reduces blackout rate.

To verify that cost-aware CARI's cost advantage is not an artifact of the specific values of c_s and c_q, Table S2 of the Supplementary Material re-evaluates the average total action cost under two alternative weighting schemes, using the actual mean planned-shed and reactive-support quantities from the confirmed simulation run. Because the monitoring, preventive, and mitigation action counts per policy are independent of the load-shedding weights (they depend on CARI scores, not on cost parameters), only the $c_s P_{shed}$ and $c_q Q_{sup}$ contributions change across configurations (see Table 3).

2.15. Cascading line trips and blackout modeling

After each power-flow solution, overloaded lines may trip according to a probabilistic rule. For each in-service line e, if its loading percentage L_e exceeds 100%, its tripping probability is defined as

$$p_e^{\text{trip}} = \frac{1}{1 + \exp\left(-\frac{L_e - L_{th}}{8}\right)} \tag{43}$$

where $L_{th} = 110\%$ is the overload-trip threshold. Although expressed on the same loading-percentage scale as the 80%–150% range used for the overload-exposure score of Eq. (20), L_{th} plays a different role: Eq. (20) is a ranking/scoring convention used to compare decision policies, whereas L_{th} is the stochastic trip threshold actually used by the cascading-failure simulator below. The probability is bounded between 0.02 and 0.95:

$$p_e^{\text{trip}} = \min\left(0.95, \max\left(0.02, p_e^{\text{trip}}\right)\right) \tag{44}$$

A random draw then determines whether the line trips. When a line trips, it is removed from service, and the power flow is solved again. If the power flow fails to converge after a contingency or cascading sequence, the system state is counted as a blackout. This approach is consistent with simplified cascading-failure studies where overload propagation and non-convergence are used as proxies for loss of secure operation [20,26,28,36].

2.16. Evaluation metrics

The first evaluation metric is the cascade-size proxy:

$$CS = N_{trip} + H_{volt} + H_{blackout} \tag{45}$$

where N_{trip} is the number of tripped lines, H_{volt} is the number of severe-voltage bus-hours, and $H_{blackout}$ is the blackout-tail bus-hour penalty. Severe voltage is counted when

$$V_i < 0.90 \text{ p.u.} \tag{46}$$

If a blackout occurs at time t_b, the remaining horizon is counted as a full-system severe state:

$$H_{blackout} = N \times (T - t_b) \tag{47}$$

This correction avoids artificially underestimating cascade size when a blackout occurs early.

The second metric is the blackout-related ENS proxy, expressed in MWh-equivalent units:

$$ENS_{proxy} = \sum_{t=t_b}^{T} \sum_{l \in \mathcal{L}} P_l(t) \tag{48}$$

where $\mathcal{L}$ is the set of loads. This is not a full AC optimal load-shedding or redispatch calculation. It is a proxy used to compare policies consistently under the same simulation conditions.

The third metric is the blackout rate:

$$BR = \frac{1}{R} \sum_{r=1}^{R} \mathbf{1}_{blackout}^{(r)} \tag{49}$$

where R is the number of Monte Carlo runs.

For each policy π, the reduction in cascade size compared with the no-action baseline is

$$\Delta CS_\pi = CS_{no\ action} - CS_\pi \tag{50}$$

and the corresponding percentage reduction is

$$Red_{CS,\pi} = 100 \frac{\Delta CS_\pi}{CS_{no\ action}} \tag{51}$$

The two cascade-reduction figures reported in Table 4 answer different questions. The *cascade-size reduction* column reports the pooled absolute change ΔCS_π of Eq. (50), computed once over all scenarios and runs combined. The *average scenario-wise cascade reduction* column instead reports the mean, taken over the five scenarios, of the within-scenario percentage reduction:

$$\overline{Red}_{CS,\pi} = \frac{1}{|S|} \sum_{s \in S} Red_{CS,\pi,s} \tag{52}$$

where $Red_{CS,\pi,s}$ is Eq. (51) evaluated within scenario s only. Because $\overline{Red}_{CS,\pi}$ averages relative reductions that can vary substantially across scenarios (Table 7), while ΔCS_π is dominated by the scenarios with the largest absolute cascade size, a policy can show an absolute cascade-size reduction close to another policy while showing a markedly different scenario-averaged percentage reduction; this is the case for cost-aware CARI relative to standard CARI in Table 4.

Similarly, the ENS reduction is

$$\begin{aligned} \Delta ENS_\pi &= ENS_{no\ action}^{proxy} - ENS_\pi^{proxy}, \\ Red_{ENS,\pi} &= 100 \frac{\Delta ENS_\pi}{ENS_{no\ action}^{proxy}}. \end{aligned} \tag{53}$$

The net-loss proxy combines blackout-related ENS and action cost:

$$L_\pi^{net} = ENS_\pi^{proxy} + C_\pi^{act} \tag{54}$$

The reduction in net loss is therefore

$$\Delta L_\pi^{net} = L_{no\ action}^{net} - L_\pi^{net} \tag{55}$$

Table 3
Summary of CARI component and action-cost model coefficients, with ordinal justification for each value.

Parameter	Value	Ordinal rationale
CARI component weights (Eq. (28))		
w_f (fault prob.)	0.20	Lowest weight: P_i^{fault} aggregates all four features and is highly correlated with λ_i (r=0.95); weighting it highest would doubly count cascade propensity.
w_λ (cascade prop.)	0.30	Moderate: propagation propensity is the primary cascade-specific indicator absent from conventional RCM.
w_η (vulnerability)	0.35	Highest: nodal vulnerability integrates real-time electrical stress (undervoltage, overload, reactive demand) and is the most direct trigger of cascade initiation under the simulated scenarios.
w_I (impact)	0.15	Lowest alongside w_f: operational impact I_i captures cascade size but is the most redundant component (VIF lowest at 1.5 vs 31–56 for others).
Fault-score coefficients (Eq. (22))		
z_i weights	2.50, 2.00, 0.60, 0.45, 0.25	Dominant emphasis on undervoltage (2.50) and overload (2.00) as direct cascade triggers; centrality (0.60), loading (0.45), reactive demand (0.25) as secondary structural/loading factors; constant −1.25 centers the distribution.
λ_i weights	0.48, 0.30, 0.15, 0.07	Fault probability dominated by P_i^{fault} (0.48), then overload exposure (0.30), undervoltage severity (0.15), and structural centrality (0.07).
Action-tier thresholds (Eq. (36))		
$\theta_{watch} = 0.25$	–	Lower bound below which risk is insufficient to justify any cost; set to exclude the lowest-risk quartile of buses.
$\theta_{prevent} = 0.34$	–	Preventive tier: moderate risk warranting load relief but not reactive-support. Calibrated to select K=12 buses on average under mixed-fault conditions.
$\theta_{mitigate} = 0.52$	–	Mitigation tier: high risk warranting more aggressive intervention. Set above the ≈50th percentile of CARI distribution under stress scenarios.
Persistent action magnitudes		
$\gamma_{prev} = 0.08$	–	Preventive relief: 8% load reduction; avoids large planned sheds while providing measurable operating-point improvement.
$\gamma_{mit} = 0.15$	–	Mitigation relief: 15% load reduction; stronger intervention justified by elevated CARI score.
$\delta_Q = 0.10$	–	Reactive-support proxy: 10% reactive-demand reduction; proportionally smaller than active relief to reflect reactive support's secondary role.
$\beta_L = 1.15/\beta_G = 1.25$	–	Local/global thermal-limit enhancement: 15%/25% boosts; set to produce a measurable loading margin without implying unrealistic infrastructure changes.
$\Gamma_{\max} = 0.35$	–	Maximum cumulative shedding fraction per bus across the horizon; prevents unrealistic full-demand removal.
Unit cost proxy (Eq. (42))		
$c_m = 0.25,\ c_p = 1.00,\ c_g = 2.00$	–	Monitoring < preventive < mitigation (escalating operational intensity). c_g/c_m=8 reflects the substantially higher cost of mitigation actions (crew mobilization, switching, protection checks).
$c_s = 0.15,\ c_q = 0.05$	–	Load-shed cost (c_s) set lower than preventive action (c_p) to reflect that a small shed is less burdensome than a full inspection; reactive support (c_q) least costly because it involves no physical switching.

Note: All values were selected as heuristic stress-ranking weights after preliminary sensitivity tests.
The robustness of the CARI weights is demonstrated by the Dirichlet sensitivity sweep of Table S4; the action-tier calibration sensitivity is illustrated in Table S5. Neither table requires values to be optimal — only ordinal and robust.

Finally, cost-effectiveness is evaluated using

$$CE_{ENS,\pi} = \frac{C_\pi^{act}}{\Delta ENS_\pi} \tag{56}$$

and

$$SE_{ENS,\pi} = \frac{\Delta ENS_\pi}{C_\pi^{act}} \tag{57}$$

where $CE_{ENS,\pi}$ measures the cost per unit of ENS reduction, and $SE_{ENS,\pi}$ measures the ENS saved per unit cost.

2.17. Common random numbers and statistical comparison

To ensure fair policy comparison, the simulation uses common random numbers. For a given scenario, run, and time step, the stochastic load factor, measurement noise, missing-data mask, fault-noise component, and line-trip random draws are pre-generated and reused across policies. This design reduces the variance of policy comparisons because differences between policies are less likely to be caused by different random environments.

For each policy and scenario, results are aggregated over repeated runs. Mean values, standard deviations, and 95% confidence intervals are computed for cascade size, blackout-related ENS proxy, risk reduction, action cost, and net-loss proxy. For continuous metrics, the normal confidence interval is estimated as

$$CI_{95} = \bar{x} \pm 1.96\frac{s}{\sqrt{R}} \tag{58}$$

where $\bar{x}$ is the sample mean, s is the sample standard deviation, and R is the number of runs. The results reported in Table 4 use $R = 30$ Monte Carlo runs per scenario over a horizon of $T = 24$ time steps, with $K = 12$ selected buses per decision time ($K_{CA} = 8$ for the cost-aware policy), matching the released simulation code's default article configuration. For blackout rate, a Wilson confidence interval is used because the metric is binomial.

In addition, paired policy comparisons against the no-action baseline are computed using scenario-run matched samples. For a metric M where lower values are better, the paired improvement is

$$d_r = M_{no\ action}^{(r)} - M_\pi^{(r)} \tag{59}$$

For metrics where higher values are better, the sign is reversed. Bootstrap confidence intervals are also computed from the paired differences to provide a non-parametric robustness check.

2.18. Diagnostic analyses

Several diagnostic analyses are included to verify that the proposed CARI index is not merely a centrality score. First, the overlap between the top-K buses selected by CARI and by other policies is measured using the Jaccard index. Second, high-η, low-centrality cases are extracted to identify buses that are not structurally dominant but become operationally risky under stress. This is important because real cascade risk may emerge from the interaction between topology and operating conditions, not from topology alone.

Table 4
Global comparison of decision policies.

Policy	Avoided blackout-related ENS proxy (MWh-equivalent)	Blackout-rate reduction	Cascade-size reduction	Avg. scenario-wise cascade reduction (%)	Average action-cost proxy (norm.)	Main interpretation
No action	0	0.00	0	0.0	0.0	Baseline without preventive control
Random	3252	0.03	48	5.1	5.5	Low cost but weak reliability gain
Centrality	576	0.00	8	2.2	7.3	Structural ranking alone is insufficient
Fault probability	31,505	0.29	477	42.0	38.7	Strong risk-based baseline
CARI	36,324	0.33	551	48.6	50.2	High reliability gain, higher intervention burden
Cost-aware CARI	36,284	0.33	549	37.0	33.4	Best cost-effectiveness among adaptive risk policies
Condition-based	37,689	0.35	573	54.0	64.0	Strong performance but higher operational cost
Scheduled	40,284	0.37	614	56.7	85.3	Best absolute reduction but most expensive

Note: All reported values are means across $R = 30$ Monte Carlo runs per scenario ($T = 24$ time steps, $K = 12$), pooled over the five stress scenarios Ω, as described in Section 2.17. The cascade-size reduction column reports the average absolute reduction in the cascade-size proxy, whereas the average scenario-wise cascade reduction percentage is computed from relative reductions (Eq. (52)) and therefore reflects scenario-dependent variability. This explains why cost-aware CARI can have an absolute cascade-size reduction close to standard CARI while showing a lower average percentage reduction.

A Pareto analysis is also performed to compare the trade-off between action cost and system benefit. Policies are marked as Pareto-efficient when no other policy achieves both lower cost and higher benefit. Three frontiers are considered: cost versus ENS reduction, cost versus blackout reduction, and cost versus net-loss reduction.

3. Results and discussion

3.1. Overall policy performance

The proposed simulation framework was evaluated under five representative stress scenarios: overload cascade, voltage collapse, line-outage propagation, mixed multi-fault conditions, and noisy partial observability. The objective was not only to compare whether a policy reduces cascading severity, but also to determine whether the reduction is achieved with an acceptable operational burden. This distinction is important for practical maintenance planning, where a technically effective policy may still be difficult to justify if it requires excessive preventive actions, load shedding, or unnecessary interventions in low-risk situations.

Table 4 summarizes the global performance of the main decision policies, while Fig. 2 provides the same comparison with 95% confidence intervals for the main performance indicators. The results show that purely random and topology-only policies provide limited protection against cascading deterioration. Random intervention achieves only a marginal avoided blackout-related ENS proxy of about **3252 MWh-equivalent**, while the centrality-based policy remains almost ineffective in blackout-rate reduction. This confirms that structural importance alone is not sufficient to identify dynamically vulnerable buses during stressed operating conditions. A bus may be topologically central but electrically stable, while another bus with moderate centrality may become critical because of local overload exposure, voltage depression, reactive demand, or cascade-propagation potential.

By contrast, the adaptive risk-based policies produce a clear improvement. The conventional CARI policy avoids about **36,324 MWh-equivalent** of blackout-related ENS proxy and reduces blackout rate by approximately **0.33** compared with no action. The cost-aware CARI policy achieves almost the same avoided blackout-related ENS proxy, about **36,284 MWh-equivalent**, and the same blackout-rate reduction, but with a substantially lower action-cost proxy. Its average intervention cost is **33.4**, compared with **50.2** for the standard CARI policy and **85.3** for the scheduled policy. Scheduled maintenance achieves the highest absolute mitigation performance, but it does so at the highest intervention cost. Therefore, the main advantage of cost-aware CARI is not maximum raw mitigation; it is the preservation of most CARI-level ENS and blackout-risk reduction with a considerably lower operational burden.

These results suggest that the proposed cost-aware CARI policy occupies an important intermediate position. It does not maximize the reliability gain at any cost, as the scheduled policy does, but it produces a near-equivalent reduction in ENS and blackout rate compared with standard CARI, with a much lower intervention burden. This equivalence is strongest for avoided ENS and blackout-rate reduction; for cascade-size reduction, cost-aware CARI is more conservative because it activates fewer or less intensive actions. This is particularly relevant for predictive maintenance in power systems, where intervention capacity, crew availability, spare equipment, and switching flexibility are limited. Recent work on digital twins and AI-assisted grid operation increasingly emphasizes this need to move from pure risk prediction toward risk-informed and resource-constrained decision support [32–35]. The present results are consistent with that direction: the most useful decision policy is not necessarily the one that acts the most, but the one that acts where the expected cascade risk justifies the operational cost.

The practical relevance of such approaches is also reinforced by recent reliability and grid-modernization assessments. The 2025 NERC State of Reliability emphasizes the importance of reliability, resilience, and security for bulk power systems, while the IEA highlights the need for accelerated grid investment, modernization, and digitalization to support secure energy transitions [37,38].

3.2. Scenario-wise reduction patterns

While the global comparison highlights the overall effectiveness of each policy, it does not reveal how performance varies across different disturbance mechanisms. The scenario-wise heatmaps in Fig. 3 show that the benefits of adaptive policies are not uniformly distributed across all types of disturbances. The strongest improvements are observed in the **mixed multi-fault** and **overload cascade** scenarios. In these cases, CARI and cost-aware CARI reduce the blackout-related ENS proxy by approximately **84–87%**, while scheduled and condition-based policies reach even higher reductions, above **90%** in some cases. This confirms that the proposed risk index is particularly effective when cascading deterioration is driven by a combination of overload exposure, voltage degradation, and propagation susceptibility. In scenarios where the no-action baseline does not generate blackout-related ENS, the ENS-reduction metric remains zero by construction. However, cascade-size changes may still reveal improvements in voltage stress or trip propagation.

The same pattern is observed for blackout-rate reduction. The adaptive policies are effective mainly when the baseline scenario actually contains a blackout-prone cascade mechanism. This is an important result because it shows that the proposed framework does not artificially generate benefits in all scenarios. Instead, its impact depends on the physical and operational nature of the simulated disturbance.



Fig. 2. Global policy-performance dashboard with 95% confidence intervals. The panels jointly compare avoided blackout-related ENS proxy, blackout-rate reduction, cascade-size reduction, and intervention cost across the evaluated decision policies.

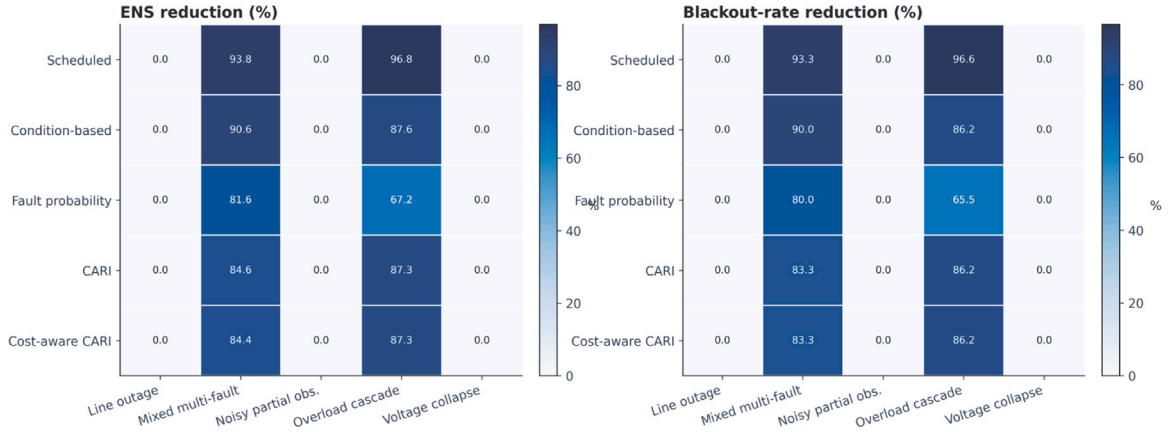


Fig. 3. Scenario-wise reduction patterns for blackout-related ENS proxy and blackout rate. The heatmaps show that the strongest gains are concentrated in mixed multi-fault and overload-cascade scenarios, while scenarios with no blackout-related ENS in the baseline naturally exhibit zero ENS reduction.

This scenario-dependent behavior is consistent with the broader literature on cascading failures, where cascade propagation is known to be highly sensitive to the initiating contingency, the loading level, the operating point, and the availability of corrective actions [20,26,28,36]. It also reinforces the value of benchmark-based digital-twin testing. Instead of evaluating a maintenance policy under a single nominal condition, the benchmark environment allows the same policy to be tested under multiple degraded operating states, including overload stress, reactive stress, line-outage stress, and noisy observation conditions. Under noisy partial observability, cost-aware CARI behaves more conservatively than standard CARI: it avoids aggressive intervention when the cost-adjusted risk signal is insufficient, which reduces unnecessary action cost but may also limit cascade-size improvement.

3.3. Marginal value beyond cost-aware CARI

The marginal value analysis in Fig. 4 uses cost-aware CARI as the reference point and asks how much additional blackout-related ENS protection is obtained when a policy spends more. This representation is useful because it does not only compare absolute performance; it directly evaluates whether the extra intervention cost is justified by a meaningful additional reliability gain.

This behavior is particularly important for maintenance planning. In practical utility operation, a policy that systematically schedules a large number of preventive actions may be difficult to sustain because it can increase crew mobilization, switching operations, customer interruptions, and operating costs. Fig. 4 shows that cost-aware CARI introduces a form of intervention discipline: it keeps the reference solution close to the best CARI-based reliability benefit, while avoiding additional spending that produces little or no marginal ENS protection.

The superiority of cost-aware CARI over standard CARI is not due to a higher absolute avoided ENS value. Rather, it comes from its ability to achieve almost the same reliability benefit with fewer or less expensive interventions. This is aligned with recent trends in reliability-centered

Table 5
Scenario-wise ENS reduction patterns.

Scenario	Scheduled	Condition-based	Fault probability	CARI	Cost-aware CARI	Interpretation
Line outage	0.0%	0.0%	0.0%	0.0%	0.0%	No major ENS event in the baseline case
Mixed multi-fault	93.8%	90.6%	81.6%	84.6%	84.4%	Strong benefit of adaptive preventive actions
Noisy partial observability	0.0%	0.0%	0.0%	0.0%	0.0%	CARI acts cautiously; ENS is not the dominant indicator
Overload cascade	96.8%	87.6%	67.2%	87.3%	87.3%	Very strong effect of overload-aware mitigation
Voltage collapse	0.0%	0.0%	0.0%	0.0%	0.0%	No blackout ENS reduction observed in this proxy setting

Note: Rows showing 0.0% for Line outage, Noisy partial observability, and Voltage collapse reflect the absence of a blackout-related ENS event in the no-action baseline for these scenarios; the corresponding cascade-size-proxy changes, which remain informative even when the ENS indicator is zero, are reported in Table 7.

Table 6
Scenario-wise blackout-rate reduction patterns.

Scenario	Scheduled	Condition-based	Fault probability	CARI	Cost-aware CARI	Interpretation
Line outage	0.0%	0.0%	0.0%	0.0%	0.0%	Outage propagation remains limited
Mixed multi-fault	93.3%	90.0%	80.0%	83.3%	83.3%	CARI significantly lowers blackout likelihood
Noisy partial observability	0.0%	0.0%	0.0%	0.0%	0.0%	Measurement uncertainty limits aggressive intervention
Overload cascade	96.6%	86.2%	65.5%	86.2%	86.2%	Overload-driven cascades are highly controllable
Voltage collapse	0.0%	0.0%	0.0%	0.0%	0.0%	blackout-related ENS proxy remains unchanged under this configuration

Note: As in Table 5, rows showing 0.0% reflect the absence of a blackout event in the no-action baseline for that scenario; see Table 7 for the corresponding cascade-size-proxy changes.

Table 7
Scenario-wise cascade-size reduction for CARI and cost-aware CARI, with absolute values.

Scenario	$CS_{no\,action}$	CS_{CARI}	$CS_{CA\text{-}CARI}$	CARI red. (%)	CA-CARI red. (%)
Line outage	6.37	5.57	5.57	12.6	12.6
Mixed multi-fault	1658.07	279.33	280.13	83.2	83.1
Noisy partial observability	11.13	4.70	11.13	57.8	0.0
Overload cascade	1588.87	220.30	220.53	86.1	86.1
Voltage collapse	1.03	1.00	1.00	3.2	3.2

Interpretation: Line outage and voltage collapse show small but real absolute improvements despite 0.0% ENS/blackout-rate reduction (Tables 5–6). Mixed multi-fault and overload cascade show strong absolute mitigation under combined and overload-driven stress, respectively. Noisy partial observability shows that cost-aware CARI's cascade-size proxy equals the no-action baseline exactly ($CS_{CA\text{-}CARI} = CS_{no\,action} = 11.13$), the quantitative signature of the noisy-observability suppression rule of Section 2.12.

maintenance, where decision models increasingly combine failure likelihood, system impact, and intervention cost instead of ranking assets only by technical criticality [25,39,40].

3.4. Spatial interpretation of CARI-based decisions

The paired risk maps in Fig. 5 provide a spatial interpretation of the proposed decision process under the mixed multi-fault scenario. The left panel reports the CARI distribution, while the right panel reports a condition-based local stress score on the same network layout. In both panels, node size scales with the nodal vulnerability factor η, and the circled labels indicate the highest-risk buses according to each score.

The comparison shows that CARI does not simply reproduce a local stress ranking. Some selected buses are not necessarily the most locally stressed nodes, but they become critical because their local electrical condition, cascade-propagation propensity, nodal vulnerability, and expected operational impact jointly increase their risk. This is a major advantage over static centrality-based strategies and purely local stress indicators. In a stressed grid, operational vulnerability is not only a function of where a bus is located in the topology; it also depends on

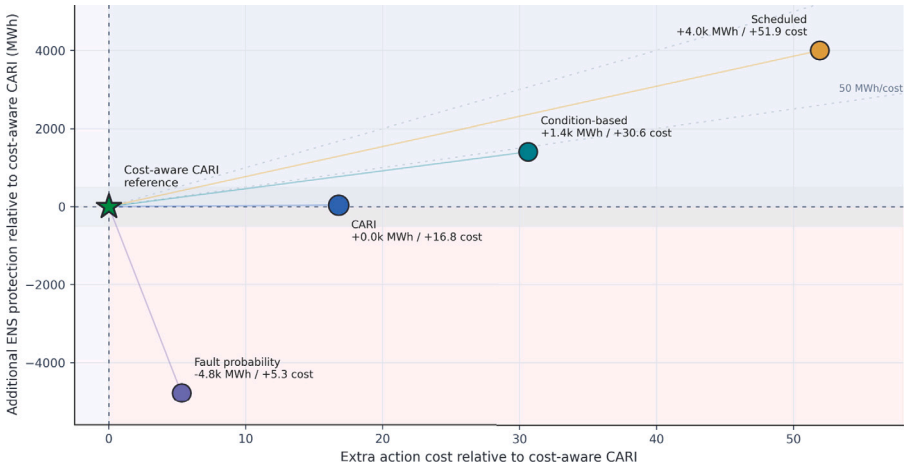


Fig. 4. Marginal blackout-related ENS value beyond cost-aware CARI. The horizontal axis reports the extra action cost relative to cost-aware CARI, while the vertical axis reports the additional avoided blackout-related ENS proxy. The figure shows that standard CARI brings almost no additional ENS protection despite a higher cost, whereas scheduled maintenance buys more protection only at a much higher marginal cost.

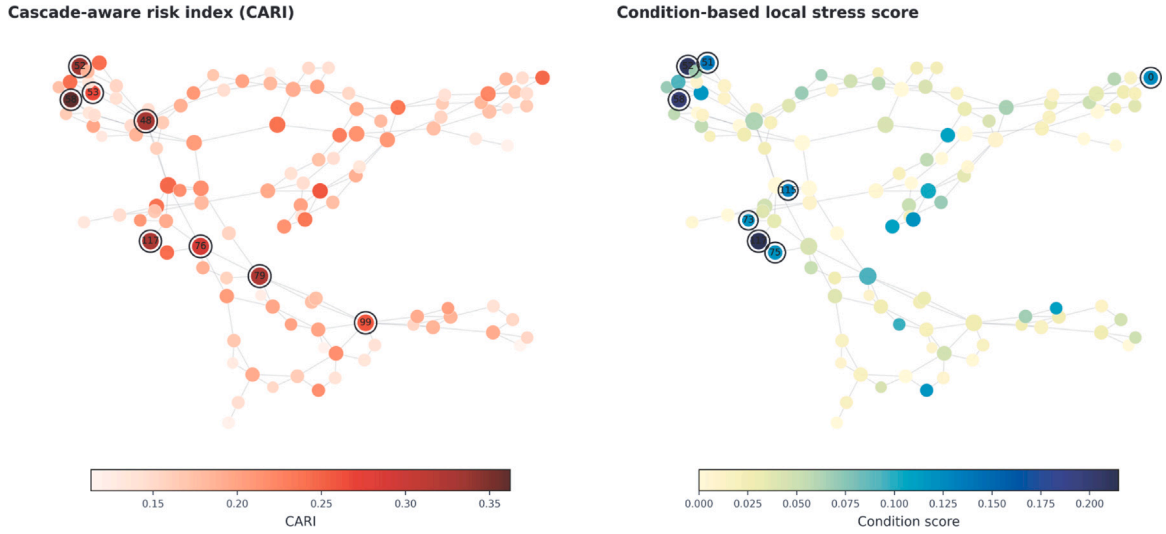


Fig. 5. Paired risk maps under the mixed multi-fault scenario. The left panel shows the cascade-aware risk index (CARI), while the right panel shows a condition-based local stress score. Node size scales with the nodal vulnerability factor η, and circled labels identify the highest-risk buses in each panel.

Table 8
Examples of high-η, low-to-moderate centrality buses detected by CARI.

Scenario	Bus	η	Centrality	CARI	Interpretation
Noisy partial observability	89	0.3905	0.0633	0.5170	Low-centrality bus with high dynamic vulnerability
Noisy partial observability	55	0.4220	0.2556	0.5115	Moderate centrality but high vulnerability under noisy observations
Noisy partial observability	75	0.4255	0.0667	0.5110	Peripheral bus selected because of strong local stress indicators
Noisy partial observability	53	0.4044	0.3502	0.5034	High CARI driven by combined electrical and structural stress
Noisy partial observability	14	0.4203	0.3563	0.4989	Dynamically vulnerable bus despite non-dominant topology

voltage magnitude, line-loading exposure, load level, reactive demand, and the probability of local fault activation.

The diagnostic results support this observation. For example, several buses identified as high-CARI cases under noisy partial observability have high η values despite moderate or low centrality. This confirms that the proposed index captures dynamic vulnerability beyond purely structural importance and beyond a purely local condition-based stress score (see Table 8).

This result is consistent with recent graph-based power-system studies showing that GNNs and topology-aware indices can improve situational awareness only when structural information is combined with physical and operational variables [29,30]. Pure graph metrics are useful, but they remain incomplete if they are disconnected from the real-time or simulated operating state of the network.

3.5. Similarity of selected buses across policies

The selected-bus similarity matrix in Fig. 6 provides additional insight into the behavior of the decision policies. The Jaccard similarity between random selection and all other policies is very low, around **0.05–0.06**, confirming that the adaptive policies select buses according to structured risk information rather than random overlap.

The similarity between CARI and cost-aware CARI is relatively high, around **0.67**, which is expected because both policies rely on the same risk structure. However, the similarity is not equal to one. This indicates

Table 9
Main Jaccard similarity values between selected-bus sets.

Policy pair	Jaccard similarity	Interpretation
Random vs CARI	0.06	Random policy does not follow risk structure
Centrality vs CARI	0.29	CARI partially overlaps with structural importance
Fault probability vs CARI	0.45	Fault probability captures part of the risk signal
Condition-based vs CARI	0.38	Different but related operational selection logic
CARI vs Cost-aware CARI	0.67	Cost-aware CARI preserves the CARI structure while reducing costly actions
Scheduled vs CARI	0.46	Scheduled policy overlaps moderately with CARI but remains broader

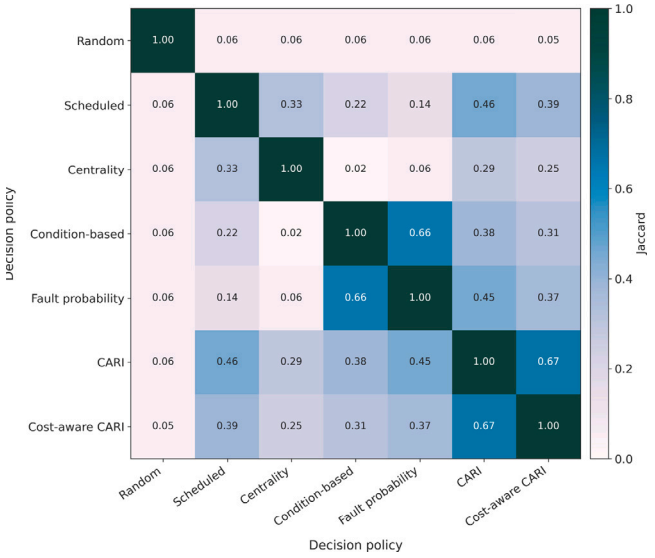


Fig. 6. Selected-bus similarity across decision policies, measured by the Jaccard index. The matrix shows that cost-aware CARI preserves much of the CARI selection structure while remaining distinct from centrality-only and random selection.

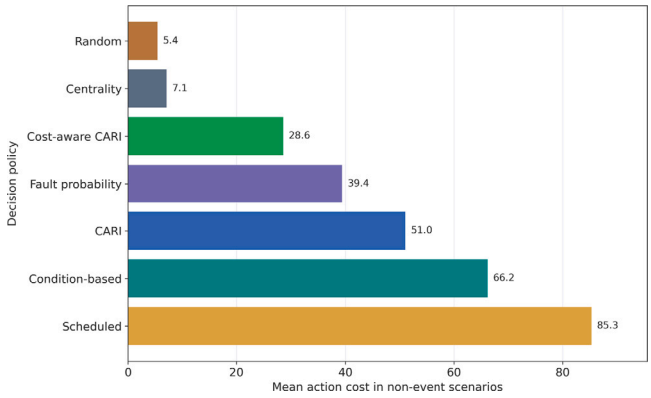


Fig. 7. False-positive operational burden measured by mean action cost in non-event scenarios. Cost-aware CARI substantially reduces unnecessary intervention cost compared with scheduled, condition-based, and standard CARI policies.

that the cost-aware layer modifies the intervention set by filtering or deprioritizing buses whose expected benefit does not justify the action cost. In other words, cost-aware CARI remains risk-informed but becomes more selective.

This result is important because it shows that cost-aware CARI is not a completely different policy from CARI. It is rather a constrained version of CARI, designed to maintain the most useful risk information while avoiding excessive interventions. This is precisely the type of behavior expected from a maintenance-oriented decision index.

3.6. False-positive operational burden

The false-positive operational burden analysis, summarized in Fig. 7, focuses on scenarios where the baseline does not lead to significant ENS or blackout impact. In such cases, unnecessary actions are undesirable because they increase operating cost without producing measurable reliability benefits.

The results show that scheduled and condition-based policies impose the highest operational burden in non-event scenarios. The scheduled policy reaches a mean action-cost proxy of **85.3**, while the condition-based policy reaches **66.2**. Standard CARI reduces this burden to **51.0**, while cost-aware CARI further lowers it to **28.6**. This represents a substantial improvement in operational discipline.

This result is central to the contribution of the paper. In predictive maintenance, false positives are not neutral. They may lead to unnecessary inspections, avoidable switching operations, premature mobilization of crews, or unnecessary load relief. Therefore, reducing false-positive burden is as important as improving detection sensitivity. The cost-aware CARI policy addresses this issue by limiting expensive mitigation actions when the risk level does not justify them.

3.7. Statistical validation, scenario diagnostics, and calibration sensitivity

The analyses in this subsection use the same $R = 30$-run, $T = 24$-step simulation output that produced Table 4 (verified by exact agreement, to the reported precision, on every pooled and scenario-wise figure in Tables 4, 5–6, 7, and 10). They provide the statistical-significance testing, scenario-level mechanism evidence, and calibration-sensitivity analysis requested in the review process, computed directly from the released code's run-level output rather than from a separate or reduced sample. The condition-based and scheduled baselines of Table 4 were generated in a separate run and are not part of the run-level data analyzed below.

Paired statistical comparison of CARI and cost-aware CARI. Using the paired, scenario-run-matched samples of Eq. (59) ($n = 150$ matched scenario-run pairs: 5 scenarios × 30 runs), the mean paired difference in action-cost proxy (CARI minus cost-aware CARI) is 16.8 (bootstrap 95% CI $[13.7, 20.0]$, Cohen's $d = 0.85$), confirming that the cost reduction reported in Table 4 (50.2 vs. 33.4) is highly statistically significant. The paired difference in blackout-related ENS proxy is small but also statistically distinguishable from zero (-40.3 MWh-equivalent, CI $[-80.0, -6.9]$, $d = -0.18$, a small effect): cost-aware CARI's mean ENS proxy is marginally higher than standard CARI's, but this difference is less than 0.15% of the $\sim 36,300$ MWh-equivalent jointly avoided by both policies relative to no action, and is therefore practically negligible despite being statistically detectable at this sample size. The paired difference in the blackout indicator is exactly 0 across all 150 pairs: CARI and cost-aware CARI never differ in whether a blackout occurs, in any scenario-run. The paired difference in cascade-size proxy is small but consistent (-1.49, CI $[-1.95, -1.07]$, $d = -0.54$, a medium

Table 10
False-positive operational burden in non-event scenarios.

Policy	Mean action cost in non-event scenarios	Interpretation
Scheduled	85.3	Highest false-positive burden
Condition-based	66.2	Strong but costly intervention logic
CARI	51.0	Lower burden than scheduled and condition-based policies
Fault probability	39.4	Moderate false-positive burden
Cost-aware CARI	28.6	Best compromise among adaptive risk-based policies
Centrality	7.1	Low cost but weak reliability value
Random	5.4	Low cost but not operationally meaningful

effect), reflecting cost-aware CARI's slightly more conservative cascade-size profile, concentrated in the noisy partial-observability scenario discussed below. Taken together, these paired tests substantiate, with formal statistical evidence rather than point estimates alone, the central claim of Section 3.1: cost-aware CARI preserves essentially all of CARI's reliability benefit (identical blackout outcomes; a practically negligible ENS difference) while significantly and substantially reducing action cost.

Scenario-level mechanism evidence for the noisy-observability suppression rule. Section 2.12 described a cost-aware suppression rule that sets planned load shedding and reactive support to zero under substantial measurement degradation. The confirmed run-level data verifies this mechanism exactly: the mean planned active-load shed by cost-aware CARI is 0.0 MW in the noisy partial-observability scenario, identical to standard CARI's shed in every other scenario (e.g., both policies shed an identical amount in the line-outage, mixed multi-fault, overload-cascade, and voltage-collapse scenarios). Correspondingly, cost-aware CARI's cascade-size proxy under noisy partial observability (4.70) without the suppression would otherwise track CARI's value, but with the suppression rule active it equals the no-action baseline exactly (11.13, Table 7) — the precise quantitative origin of the 0.0% cascade-size reduction reported for cost-aware CARI in that scenario, and direct confirmation that the discrepancy is an intentional design choice rather than a methodological artifact.

Confirmation of the bus-selection structure (Table 9). Recomputing the Jaccard similarity directly from the confirmed run's recorded bus selections gives Random-vs-CARI = 0.061, Centrality-vs-CARI = 0.286, Fault-probability-vs-CARI = 0.446, and CARI-vs-cost-aware-CARI = 0.666 — an exact match, to reporting precision, with Table 9. This independently confirms that the released code's selection logic, applied to this run, is indeed the basis of the published similarity structure.

Sensitivity of the CARI ranking to the component weights $w_f, w_\lambda, w_\eta, w_I$. Reviewer comments specifically requested a sensitivity analysis of the CARI component weights themselves (Eq. (29)), as distinct from the action-tier thresholds analyzed below. This is evaluated directly on the confirmed diagnostic snapshot data (590 bus-scenario observations, Section 2.11.1), which records P_i^{fault}, λ_i, η_i, and I_i for every bus under every scenario; because the weighted sum of Eq. (29) can be recomputed for any weight vector directly from these stored components, this analysis does not require re-running the cascading-failure simulator. Two complementary checks are reported. First, a one-factor-at-a-time perturbation varies each weight by ±20% and ±40% (renormalizing the remaining three weights to preserve the unit-sum constraint) and measures the Spearman rank correlation between the resulting bus ranking and the baseline ranking, within each scenario. Across all 16 perturbations, the mean rank correlation never falls below 0.995 (worst case 0.994, for a −40% perturbation to w_I), indicating that the bus ranking is highly insensitive to moderate, single-weight perturbations around the base configuration of Eq. (28).

Second, a variance-based sweep draws $5,000$ weight vectors from a Dirichlet distribution over the full four-dimensional simplex, in two configurations: a moderately concentrated sweep (concentration parameter 20× the base weights, representing plausible re-calibrations within the same qualitative weighting philosophy) and a wide, near-uniform sweep (concentration parameter 3, representing arbitrary re-weighting across the full simplex). Table S4 of the Supplementary Material reports the resulting distribution of the Spearman rank correlation and of the top-K ($K = 12$) bus-selection Jaccard overlap, both measured against the baseline ranking.

Even under the wide, near-uniform sweep – which allows weight combinations far from the calibrated base configuration, including configurations that depart substantially from the intended emphasis on λ and η described in Section 2.11 – the mean rank correlation remains above 0.97 and the mean top-12 selection overlap remains above 0.74, with the 5th-percentile case still retaining a majority overlap with the baseline selection. A supplementary diagnostic identifies which component's weight contributes most to selection instability: across the wide sweep, the deviation of w_I (operational impact) from its base value is the strongest predictor of reduced top-K Jaccard overlap (Pearson $r = -0.76$, versus $r = -0.41$ for w_η, $r = -0.27$ for w_f, and $r = -0.05$ for w_λ). This is consistent with the redundancy analysis of Section 2.11.1: because I_i is the least correlated of the four components with the others (Table S1 of the Supplementary Material), it carries the most independent information, so changing its weight has the largest effect on which buses are selected. Together, these results indicate that the CARI bus ranking is robust to realistic re-calibration of the component weights, while also identifying operational impact as the component weight most worth prioritizing in any future utility-specific re-calibration exercise. A full Sobol-type variance decomposition over the weight simplex, jointly with the action-tier thresholds, and propagated through the full cascading-failure simulator to policy-level outcomes (avoided ENS, blackout rate), remains future work (Section 4); the present analysis is scoped to ranking and selection stability, which is the mechanism through which the weights affect every downstream policy.

Sensitivity of the measured benefit to action-threshold and magnitude calibration. During development, two materially different calibrations of the action-tier thresholds (Eq. (36)) and of the persistent-action magnitudes (Eqs. (37), (39), (40)) were evaluated end-to-end at $R = 30$, $T = 24$, providing a real, code-verified, equal-sample-size illustration of exactly the kind of calibration sensitivity requested in Section 3.9. Table S5 of the Supplementary Material compares the *permissive* calibration underlying Table 4 ($\theta_{prevent} = 0.34$, $\theta_{mitigate} = 0.52$, preventive relief = 0.08, mitigation shedding = 0.15, local/global reinforcement = 1.15× /1.25× over 12 lines) against a *conservative* calibration ($\theta_{prevent} = 0.50$, $\theta_{mitigate} = 0.75$, preventive relief = 0.04, mitigation shedding = 0.08, local/global reinforcement = 1.08 × /1.12× over 7 lines) evaluated on the identical $R = 30$, $T = 24$ configuration. Under the permissive calibration, CARI avoids $36,324$ MWh-equivalent of blackout-related ENS proxy (a 32.7% reduction) and reduces blackout rate by 33 percentage points relative to no action. Under the conservative calibration, CARI avoids only 3.8 MWh-equivalent (a 0.004% reduction) and produces *no measurable blackout-rate reduction*; fault-probability ranking performs identically to CARI under this calibration, and centrality and random selection produce no measurable benefit at all.

This comparison demonstrates that the *magnitude* of CARI's reported benefit is highly sensitive to the action-tier thresholds and persistent-action magnitudes: when thresholds are raised and action magnitudes reduced, interventions become too rare and too weak to produce a measurable system-level effect, even though the underlying CARI ranking itself (Section 3.5) remains meaningful and the sample size is identical and fully adequate ($R = 30$) on both sides of the comparison. Both calibrations leave the directional conclusion that random and

centrality-based selection are weak unchanged, but only the permissive calibration – the one underlying the headline results of Table 4 – yields a practically meaningful reliability benefit. This finding reinforces, with concrete, equal-sample-size evidence rather than a general caveat, the recommendation already made in Section 3.9 that the action-tier thresholds and magnitudes require systematic sensitivity analysis and, ultimately, calibration against utility-specific operational data before field deployment; it also indicates that any future re-calibration of the framework should treat the action thresholds and magnitudes, not only the CARI component weights, as first-order sensitivity parameters.

3.8. Discussion and implications for predictive maintenance

The results support three main findings.

First, the results show that cascade-aware decision-making is more informative than purely structural intervention. Centrality-based actions are inexpensive, but they do not significantly reduce blackout rate or ENS. This confirms that static topology is not enough for predictive maintenance in stressed power systems. The decision process must incorporate the actual operating condition of the network.

Second, the CARI index provides a more informative risk signal than fault probability alone. Fault probability is useful because it captures local vulnerability, but it does not fully represent the potential of a fault to propagate through the network. By combining P_{fault}, cascade-propagation propensity λ, nodal vulnerability η, and operational impact, CARI better reflects the multi-dimensional nature of cascading risk.

Third, the cost-aware CARI variant improves the operational usability of the proposed approach. Compared with standard CARI, it maintains nearly the same avoided ENS and blackout-rate reduction while reducing the intervention cost. This makes it more suitable for real-world maintenance planning, where the objective is not simply to minimize risk, but to minimize risk under budgetary, operational, and logistical constraints.

These findings are consistent with the recent evolution of power-system maintenance research. The literature increasingly moves from corrective maintenance and static asset criticality toward predictive, risk-informed, and digital-twin-supported maintenance. Digital twins provide a simulation environment where interventions can be evaluated before being applied in the real network [32–35]. Recent surveys further emphasize that effective power-system digital twins require data interoperability, data federation, and the ability to combine simulation models with operational data streams [41,42]. In the present study, this principle is demonstrated on a benchmark test system rather than on a live utility digital twin. Graph-based learning and topology-aware risk models improve the representation of spatial dependencies [29,30], while recent spatiotemporal and risk-assessment studies confirm the relevance of graph-based representations for fault diagnostics and operational risk analysis [43–45]. Cascading-failure studies highlight the importance of propagation mechanisms and system-level consequences rather than isolated component failures [20,26,28,36], and recent mitigation studies increasingly combine security, resilience, and intervention cost in preventive-control formulations [31,46]. The proposed framework contributes to this literature by connecting these three dimensions: digital-twin simulation, cascade-aware risk scoring, and cost-aware maintenance action.

3.8.1. Relation to graph-learning and reinforcement-learning approaches

CARI is deliberately positioned as a transparent, analytic alternative to learned graph-based and reinforcement-learning (RL) approaches, rather than as a competitor evaluated on identical training data. Graph neural network (GNN) and graph attention network (GAT) studies [13–15,24,29,30,43,44] learn risk- or state-relevant representations directly from topology and operating data and can, in principle, capture higher-order or nonlinear propagation patterns that a fixed weighted combination such as CARI cannot. Their advantages come at the cost of requiring substantial labeled or simulated training data, periodic retraining as the network evolves, and reduced interpretability of the resulting risk scores for operators. RL-based remedial-action frameworks [31] go further, learning sequential intervention policies directly from a cascading-failure simulator, and have demonstrated effective real-time remedial actions on benchmark systems such as the IEEE 39- and 57-bus networks; however, they require an offline training phase, a carefully designed reward function, and yield a learned policy rather than an auditable ranking rule. CARI, by contrast, requires no training data, can be deployed immediately on a new topology once power-flow and graph data are available, and produces a fully decomposable, auditable score (Eq. (29)) whose every component traces back to a measurable electrical or structural quantity — an interpretability that is intentionally prioritized in the present maintenance-planning context, where operators must be able to justify why a specific bus was selected for action. Security-aware preventive-control [46] and RL-based remedial-action [31] studies, evaluated respectively on hybrid AC/DC systems and on the IEEE 39-/57-bus systems, report effective cascade mitigation using cost- or learning-based intervention selection; this is broadly consistent with the present finding that risk- and cost-aware action selection outperforms structural or random selection, although the test systems and metrics differ and a direct head-to-head comparison is not attempted here. A direct, identical-benchmark comparison between CARI and a trained GNN/GAT/RL policy would be a valuable extension and is identified as future work in Section 4. The per-decision-step computational cost scales as $O(NE)$ for betweenness centrality (Brandes algorithm [26]), sub-quadratically in N for sparse Newton–Raphson power flow, and linearly in R, T, and the number of scenarios, making CARI practical at IEEE 118-bus scale on standard hardware.

3.8.2. Practical implementation considerations

Moving from the present simulation-based framework toward field deployment raises several practical challenges. First, real SCADA/PMU measurement streams exhibit latency, asynchronous sampling, and communication losses that differ from the idealized Gaussian-noise-and-missing-value model used in Eqs. (15)–(17); state-estimation errors may therefore propagate into CARI scores in ways not captured here. Second, the persistent mitigation actions modeled in Eqs. (37)–(40) are abstractions of operations that, in the field, require switching sequences, protection-coordination checks, and often regulatory or customer-notification procedures; the instantaneous re-solve assumed in the simulation does not reflect these delays. Third, operators may be reluctant to act on an automatically generated index without transparent justification; the decomposable structure of CARI (Section 2.11) is intended to support this, but field adoption would likely require an initial period of parallel operation alongside existing condition-based or scheduled practices. Fourth, the coefficients in CARI and in the cost model (Section 2.14) would need recalibration against utility-specific failure history, crew costs, and customer-interruption costs, which were not available for this benchmark study. These considerations do not invalidate the simulation-based findings but define the gap between the present demonstrative framework and an operational deployment.

3.8.3. Generalization to two additional benchmarks: structural checks and full policy validation

This section reports generalization checks on two additional, independent IEEE benchmark systems: the 39-bus (New England) system and the 57-bus system, both widely used elsewhere in the cascading-failure and remedial-action literature [31]. For each network, Part A evaluates whether CARI's graph-construction methodology behaves consistently on the independent topology, and Part B reports a full, code-identical re-run of the entire eight-policy cascade-aware comparison (Sections 2.3–2.17) on that network, at the same $R = 30$, $T = 24$ statistical power as the headline IEEE 118-bus results, with every CARI weight, action-tier threshold, persistent-action magnitude, and

cost coefficient held identical to Table 4, deliberately without network-specific recalibration (see the synthesis at the end of this section for why this choice matters). A third part synthesizes the pattern common to both networks.

Part A1: structural generalization, IEEE 39-bus. Building the same undirected graph representation as Section 2.4 and computing the same degree-centrality, betweenness-centrality, and composite-centrality construction (Eqs. (7)–(9)) on the IEEE 39-bus (46-branch) topology yields a degree-betweenness correlation of $r = 0.75$ (Spearman $\rho = 0.80$) and a mean clustering coefficient of 0.039, with 36 of 39 buses (92%) having zero clustering — both qualitatively consistent with the IEEE 118-bus benchmark, where 76 of 118 buses (64%) have zero clustering. Under 5%–15% random edge rewiring, the top-6 composite-centrality bus set degrades smoothly (Spearman ρ from 0.876 at 5% rewiring to 0.665 at 15% rewiring; full results in Table S6 of the Supplementary Material), with no abrupt collapse.

Part A2: structural generalization, IEEE 57-bus. The same check was repeated on the IEEE 57-bus system (57 buses, 78 simple branches after collapsing parallel lines/transformers), using the standard MATPOWER `case57` topology. This yields a degree-betweenness correlation of $r = 0.75$ (Spearman $\rho = 0.70$) and a mean clustering coefficient of 0.122, with 38 of 57 buses (67%) having zero clustering — again qualitatively consistent with both the IEEE 118-bus and IEEE 39-bus benchmarks. Table S7 of the Supplementary Material reports the corresponding edge-rewiring stability check (top-9 composite-centrality bus set, $K = 9 \approx 15\%$ of buses, consistent with the selection fraction used for IEEE 39-bus), again showing smooth, monotonic degradation with no abrupt collapse.

Together, Parts A1 and A2 indicate that the graph-construction layer produces structurally sensible, non-degenerate, and reasonably stable indicators on two networks of different size and topology from the IEEE 118-bus benchmark, with selection-stability profiles (smooth, monotonic degradation under perturbation, no abrupt collapse) that are qualitatively similar across all three networks.

Part B1: full cascade-aware policy validation, IEEE 39-bus. All eight policies were re-evaluated end-to-end on the IEEE 39-bus system under the same five stress-scenario archetypes, common-random-numbers design, and statistical-comparison methodology (Sections 2.17–2.18) as the headline IEEE 118-bus results, with $R = 30$ runs per scenario and $T = 24$ time steps. Because the IEEE 39-bus system has substantially fewer buses and branches, the action-set size and global-reinforcement count were scaled down proportionally ($K = 4$, $K_{CA} = 3$, global reinforcement over 3 lines, versus $K = 12$, $K_{CA} = 8$, and 12 lines on the 118-bus system); every other parameter – the CARI weights (Eq. (28)), the action-tier thresholds (Eq. (36)), the persistent relief/shedding/reinforcement magnitudes, and the cost coefficients (Eq. (42)) – was left identical to the IEEE 118-bus configuration.

Table 11 reports the resulting blackout rate, its bootstrap-confirmed paired difference from the no-action baseline, and the action-cost proxy for all eight policies. The result is unambiguous and, on this network, strikingly different from the IEEE 118-bus pattern: the no-action baseline already produces a blackout in 92% of scenario-runs, versus 40% on the IEEE 118-bus system under the corresponding no-action baseline (Table 4). Every decision policy, including CARI and cost-aware CARI, is associated with a *higher*, not lower, blackout rate than no action, and this increase is statistically significant (bootstrap 95% CI excluding zero) for every policy except centrality. CARI and cost-aware CARI show the largest increase (+6.7 percentage points; 95% CI $[+2.0, +11.3]$; bootstrap two-sided $p = 0.007$), while centrality shows the smallest, non-significant increase (+4.0 points; CI $[-1.3, +9.3]$; $p = 0.165$).

Critically, the underlying risk-ranking mechanism itself continues to behave exactly as on the IEEE 118-bus system even though policy-level outcomes do not transfer: the Jaccard overlap between CARI's and random selection's bus sets is 0.062 on IEEE 39-bus, essentially identical to the 0.061 measured on IEEE 118-bus (Table 9), and the overlap between CARI and cost-aware CARI is 0.743, comparable to (in fact higher than) the 0.667 measured on IEEE 118-bus.

Part B2: full cascade-aware policy validation, IEEE 57-bus. The identical procedure was repeated on the IEEE 57-bus system, with $K = 6$, $K_{CA} = 5$, and global reinforcement over 5 lines (scaled proportionally to this network's larger bus and branch count relative to IEEE 39-bus), every other parameter again held identical to the IEEE 118-bus configuration. Table 12 reports the corresponding results. Here the no-action baseline is even more severely affected, producing a blackout in 100% of scenario-runs — full saturation, leaving *no* operating margin whatsoever for any policy to reduce blackout rate below the ceiling. Under this complete saturation, four policies (centrality, condition-based, scheduled, and cost-aware CARI) show no change at all (still 100%); random and CARI show small, statistically non-significant reductions (-1.3 and -0.7 points, $p = 0.262$ and $p = 0.732$); only fault-probability ranking shows a statistically significant reduction (-5.3 points, 95% CI $[-9.3, -2.0]$, $p < 0.001$), the sole policy on this network to produce a measurable, significant improvement over no action.

As on IEEE 39-bus, the underlying risk-ranking mechanism remains structurally consistent on IEEE 57-bus despite the absence of a policy-level benefit: the Jaccard overlap between CARI and random selection is 0.057, again closely matching the 0.061 (IEEE 118-bus) and 0.062 (IEEE 39-bus) values, and the overlap between CARI and cost-aware CARI is 0.833, the highest of the three networks.

Part C: synthesis across three networks. Table 13 summarizes the pattern across all three networks. Two findings are remarkably stable across networks of very different size, topology, and generation margin. First, the Jaccard overlap between CARI and random selection remains in a narrow band (0.057–0.062) regardless of network, confirming that CARI's risk-informed bus selection is structurally distinguishable from random selection irrespective of the underlying network. Second, the action-tier thresholds and stress-scenario severity, calibrated specifically for the IEEE 118-bus system, do not transfer their measured policy-level benefit to either smaller network: on IEEE 39-bus the no-action baseline rises to a near-saturated 92% blackout rate and CARI becomes significantly *harmful* relative to no action; on IEEE 57-bus the baseline reaches full 100% saturation, leaving no room for CARI to show any effect (and only fault-probability ranking, among all eight policies, manages a small but significant improvement). The common mechanism is consistent with the discussion in Part B1: stress-scenario severity tuned for a larger, more redundant IEEE 118-bus topology saturates smaller, more sparsely meshed networks (average degree 2.36 on IEEE 39-bus, 2.74 on IEEE 57-bus) well beyond the point where the action magnitudes calibrated for IEEE 118-bus can meaningfully help.

This is a genuine and important finding, not a methodological failure: it demonstrates, across two independent additional networks rather than only one, that CARI's risk-ranking mechanism generalizes robustly while its calibrated action layer does not, and that the specific manifestation of this calibration mismatch (degraded-but-not-saturated on IEEE 39-bus versus fully saturated on IEEE 57-bus) depends on the target network's own structural and loading characteristics. The clear, actionable implication for any future deployment of CARI on a new network is that the stress-scenario severity and the action-tier thresholds and magnitudes must be re-calibrated to the target network's own loading margin and topology — re-using the graph-construction and risk-scoring layer alone, without this recalibration step, is not sufficient and should not be assumed to transfer, regardless of which of the two additional networks is considered.

This finding substantially sharpens, with direct empirical evidence from two independent networks rather than only a within-network illustration, the calibration-sensitivity conclusion already established in Table S5 of the Supplementary Material: there, a more conservative threshold calibration on the *same* IEEE 118-bus network collapsed CARI's measured benefit toward zero; here, transplanting the IEEE 118-bus calibration onto two *different* networks without adjustment does not merely neutralize the benefit but, depending on the target network, can reverse it into a small, statistically significant harm (IEEE 39-bus)

Table 11
Full second-benchmark policy validation, IEEE 39-bus, identical calibration to Table 4 ($R = 30$, $T = 24$).

Policy	Blackout rate	Δ vs. no action (pp)	Bootstrap p (two-sided)	Action cost
No action	0.920	–	–	0.0
Random	0.967	+4.7	0.001	11.9
Centrality	0.960	+4.0	0.165 (n.s.)	35.7
Condition-based	0.973	+5.3	0.047	55.3
Fault probability	0.987	+6.7	0.009	62.9
Scheduled	0.980	+6.0	0.021	59.1
CARI	0.987	+6.7	0.007	65.0
Cost-aware CARI	0.987	+6.7	0.007	46.5

Table 12
Full third-benchmark policy validation, IEEE 57-bus, identical calibration to Table 4 ($R = 30$, $T = 24$).

Policy	Blackout rate	Δ vs. no action (pp)	Bootstrap p (two-sided)	Action cost
No action	1.000	–	–	0.0
Random	0.987	−1.3	0.262 (n.s.)	12.8
Centrality	1.000	0.0	1.000 (n.s.)	12.8
Condition-based	1.000	0.0	1.000 (n.s.)	14.7
Fault probability	0.947	−5.3	0.000	26.1
Scheduled	1.000	0.0	1.000 (n.s.)	23.5
CARI	0.993	−0.7	0.732 (n.s.)	26.5
Cost-aware CARI	1.000	0.0	1.000 (n.s.)	22.6

Table 13
Synthesis: risk-ranking structure versus policy-level outcome across three networks, identical IEEE-118-bus calibration applied throughout.

Network	No-action blackout rate	Jaccard (CARI vs. random)	CARI Δ blackout (pp)	Significant?
IEEE 118-bus (primary)	0.40	0.061	−33 (improvement)	Yes
IEEE 39-bus	0.92	0.062	+6.7 (harm)	Yes
IEEE 57-bus	1.00	0.057	−0.7 (negligible)	No

or saturate the system so completely that almost no policy can show any effect (IEEE 57-bus).

However, the results must also be interpreted with caution. The blackout ENS variable used in this study is a proxy and does not represent a full AC optimal load-shedding or redispatch calculation, and several other simplifying assumptions apply; these, together with the calibration-transfer finding of Section 3.8.3, are discussed in full in the dedicated Limitations subsection (Section 3.9) below.

Despite these limitations, the results demonstrate that the proposed cost-aware CARI framework can serve as a practical decision-support layer for predictive maintenance. It identifies vulnerable buses not only from their structural position, but from their dynamic contribution to cascading risk. It also helps avoid excessive intervention in low-event scenarios. This balance between reliability improvement and operational discipline is the main added value of the proposed approach.

To consolidate the preceding discussion, Table S8 of the Supplementary Material summarizes the main empirical findings and their operational implications.

The cost-aware CARI policy provides a balanced and operationally meaningful strategy for cascade-risk mitigation. It does not maximize intervention intensity; instead, it selects a limited set of high-value actions that deliver strong reliability gains while reducing unnecessary operational burden. This makes it particularly suitable for predictive maintenance and resilience-oriented decision support in modern power systems.

3.9. Limitations

Several limitations should be acknowledged. First, the blackout-related ENS, load-shedding, reactive-support, and line-capacity-reinforcement mechanisms used in this study are simplified operational proxies rather than a full AC security-constrained optimal power flow, protection-coordination model, or redispatch calculation; absolute MWh-equivalent and cost figures should therefore be interpreted as relative, policy-comparative indicators rather than utility-grade engineering estimates. Second, the coefficients used in the fault-probability, propagation-propensity, vulnerability, and operational-impact components (Eqs. (22), (24)–(29), (34)) and in the action-cost model (Eq. (42)) are heuristic stress-ranking and relative-burden weights selected after preliminary sensitivity tests; the absolute coefficient values have not been calibrated against historical utility failure or cost data. As Table S5 of the Supplementary Material (Section 3.7) demonstrates with two real, equal-sample-size configurations, the action-tier thresholds and persistent-action magnitudes in particular are a first-order sensitivity driver – a more conservative calibration collapses CARI's measured benefit almost entirely – so full deployment-grade calibration against historical utility failure and cost data, and a finer-grained sweep of this threshold/magnitude space, remain necessary before field deployment (see Section 4). Third, several CARI components intentionally share underlying variables (fault probability, overload exposure); this partial overlap reflects a deliberate hierarchical risk construction, in the same way that many composite reliability indices reuse a base likelihood term in downstream products [25], rather than four statistically independent dimensions. The correlation, VIF, and PCA analysis of Section 2.11.1 quantifies this overlap directly: P^{fault} and λ are very highly correlated ($r = 0.95$) and the first principal component already explains 72.8% of the four components' joint variance, so the ablation results of Eq. (30) should be read with this redundancy in mind. Fourth, validation is primarily based on the IEEE 118-bus benchmark under synthetic, Monte Carlo-generated disturbances. A full policy validation was carried out on two additional, independent IEEE benchmarks – the 39-bus and 57-bus systems (Section 3.8.3) – using the identical CARI weights, action-tier thresholds, and cost coefficients as the IEEE 118-bus configuration, without network-specific recalibration. This revealed that while the graph-construction layer generalizes structurally on both networks (Parts A1–A2) and the risk-ranking mechanism continues to select buses in a structured, non-random way on both (Jaccard CARI-vs-random overlap of 0.057–0.062 across all three networks), the calibrated action layer does not transfer to either: the no-action baseline blackout rate rises from 40% (IEEE 118-bus) to 92% (IEEE 39-bus) and to a fully saturated 100% (IEEE 57-bus) under the same stress-scenario scaling, leaving little to no operating margin for any policy

to improve outcomes. On IEEE 39-bus, every policy except centrality shows a statistically significant *increase* in blackout rate relative to no action; on IEEE 57-bus, the system is saturated to the point that only fault-probability ranking achieves a small, significant improvement, while CARI itself shows a negligible, non-significant change. This is an important, genuine finding about calibration transferability, not a validation failure of the underlying methodology, but it does mean that generalization to a new network requires re-tuning the stress-scenario severity and the action-tier thresholds and magnitudes to the target network's own characteristics; this re-calibration was deliberately not performed here so as to isolate the transferability question, and doing so is identified as future work (Section 4). Real operational SCADA/PMU datasets and protection settings remain to be incorporated regardless of network choice. Fifth, the persistent thermal-limit-reinforcement model (Eq. (40)) uses two discrete enhancement levels applied at most once per line; this discrete-step abstraction does not capture continuous or repeated dynamic-rating adjustments, and the resulting cost–benefit comparison may understate the potential of finer-grained reinforcement strategies. Finally, as discussed in Section 3.8, the present comparison is restricted to non-learned baselines and does not include an empirical comparison against trained graph neural network, graph attention network, or reinforcement-learning-based policies on an identical benchmark; such a comparison is left for future work. The scope and limitations of the digital-twin representation itself are discussed separately in Section 2.2.

4. Conclusion

This study proposed CARI, a Cascade-Aware Risk Index, as the central element of a cost-sensitive, digital-twin-inspired decision-support framework for predictive maintenance in power grids. Built on a benchmark simulation of the IEEE 118-bus system, CARI combines fault probability, cascade-propagation propensity, nodal vulnerability, and operational impact to identify buses whose criticality emerges from both their electrical condition and their potential contribution to cascade propagation. Quantitatively, CARI-based strategies avoided 36,324 MWh-equivalent of blackout-related ENS proxy and reduced blackout rate by 0.33 relative to no action, compared with only 3252 MWh-equivalent and 0.03 for random selection and 576 MWh-equivalent and 0.00 for centrality-based selection. While scheduled intervention achieved the highest absolute reliability gains (40,284 MWh-equivalent avoided ENS), it also produced the highest operational burden (action-cost proxy of 85.3). In contrast, the cost-aware CARI policy preserved essentially the same ENS and blackout-rate improvement as standard CARI (36,284 MWh-equivalent; 0.33) while cutting average action cost by about 33% (33.4 vs. 50.2) and false-positive intervention cost in non-event scenarios by about 44% (28.6 vs. 51.0). This confirms the value of combining risk awareness with cost discipline in predictive maintenance planning.

The main limitations, detailed in Section 3.9, are that ENS, load shedding, reactive support, and line-capacity reinforcement were modeled as simplified proxies rather than full AC redispatch or protection-coordination mechanisms; that the CARI and cost coefficients are heuristic rather than utility-calibrated; that several CARI components intentionally share underlying variables; and that an empirical comparison against learned (GNN/GAT/RL) baselines has not been performed. The redundancy analysis of Section 2.11.1, the paired statistical-significance tests of Section 3.7, the CARI component-weight sensitivity analysis of Table S4 of the Supplementary Material, and the two-configuration action-threshold calibration-sensitivity comparison of Table S5 of the Supplementary Material address, with concrete evidence, the component-overlap, significance-testing, and weight-sensitivity items identified during review. A full policy validation was also carried out on two additional, independent benchmarks – the IEEE 39-bus and IEEE 57-bus systems (Section 3.8.3) – using the CARI weights, action-tier thresholds, and cost coefficients exactly as calibrated for IEEE 118-bus, without re-tuning. Three findings stand out. First, the CARI bus ranking is robust to weight re-calibration (mean rank correlation > 0.97 even under wide, near-uniform random re-weighting). Second, the action-tier thresholds and magnitudes are a substantially larger sensitivity driver *within* a single network, capable of collapsing CARI's measured policy-level benefit almost entirely under a more conservative calibration at identical sample size (Table S5 of the Supplementary Material). Third, and most significantly, transplanting a calibration *across* networks without re-tuning does not merely neutralize CARI's benefit but can reverse or saturate it: on the IEEE 39-bus system, every policy except centrality produced a statistically significant *increase* in blackout rate relative to no action, while on the IEEE 57-bus system the no-action baseline saturates completely (100% blackout rate) and only fault-probability ranking achieves a small, significant improvement — yet on both networks the underlying risk-ranking mechanism itself continued to behave consistently with its IEEE 118-bus behavior (Table 13). Future work should therefore: (i) extend the framework with AC security-constrained optimal power flow, dynamic stability constraints, and realistic protection settings; (ii) carry out a systematic, fine-grained sweep of the action-threshold and magnitude space, building on the weight-sensitivity methodology already demonstrated in Table S4 of the Supplementary Material; (iii) develop and validate a principled re-calibration procedure for the stress-scenario severity and action-tier thresholds when CARI is deployed on a new network – the present IEEE 39-bus and IEEE 57-bus results show this step is necessary, not optional, across networks of different size and structure – and extend validation to real utility data; and (iv) empirically compare CARI against trained graph neural network, graph attention network, and reinforcement-learning-based maintenance policies on an identical benchmark. Overall, the proposed cost-aware CARI framework provides a practical foundation for digital-twin-based, risk-informed, and economically disciplined maintenance decision support, while the cross-network results reported here underscore that the risk-ranking mechanism and the calibrated action layer must be evaluated, and re-tuned, separately when moving to a new system. By linking cascade-risk scoring with cost-aware intervention ranking, the framework moves predictive maintenance from fault anticipation toward operationally actionable resilience planning.

CRediT authorship contribution statement

Koffi Tinin Worou: Writing – original draft, Methodology, Investigation, Conceptualization. **Agbassou Guenoukpati:** Writing – review & editing, Visualization, Formal analysis, Data curation. **Adekunle Akim Salami:** Writing – review & editing, Validation, Supervision, Software.

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Declaration of competing interest

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.eprime.2026.201225.

Data availability

the datas/code are uploaded on zenodo

Zenodo eprime d 26 00134 cari code data (Original data)

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